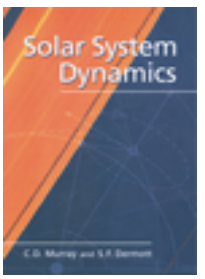
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Solar System Dynamics

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Chapter

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9
Chaos and Long-Term Evolution

Take but degree away, untune that string,
And hark what discord follows. Each thing meets
In mere oppugnancy.

William Shakespeare, *Troilus and Cressida, I, iii*

9.1 Introduction

In this book we have derived a number of equations of motion to study the rotational and orbital motion of solar system objects. These equations have described either conservative systems, such as the two- and three-body problems, or dissipative systems, such as the equations governing tidal evolution or the dynamical effects of drag forces. However, all have a common characteristic: They describe systems that are *deterministic.* This means that the current state of the system allows us to calculate its past and future state providing we know all the forces that are acting on it. In the case of the two-body problem we were able to solve the equations of motion and calculate the behaviour of the system at all past and future times. A complete analytical solution was not possible in the case of the three-body problem and we had to rely on numerical solutions if we wanted to follow the orbital evolution of a test particle. However, there was an implicit assumption that, given the initial state of the system, we should be able to calculate its future state by obtaining solutions of the equations of motion. Unfortunately this assumption is not valid for some of the systems we have investigated and this is because of the phenomenon called *chaos.*

In Chapter 7 we saw how the simple assumptions of secular perturbation theory allowed us to derive an analytical solution to the problem of the mutual interaction of N masses. Although we gained no information about the positions of bodies on their orbits we could at least calculate the other orbital elements at any time, past or present, from knowledge of the current values. The solution of the secular problem by Laplace allowed him to make assertions about the

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long-term stability of the solar system. Laplace himself believed in a deterministic universe where once the laws of nature had been discovered, it would only be a matter of knowing all the initial conditions and solving the appropriate equations; then everything about the system would be known. We now know that this view is false.

In the late nineteenth century Henri Poincaré, now acknowledged as the founder of the science of nonlinear dynamics, began a study of the mathematics of the three-body problem. This seminal work, eventually published in his *Les Méthodes Nouvelles de la Mécanique Céleste,* (Poincaré 1892, 1893, 1899), hinted at the complicated nature of the motion that can arise in solutions to the problem. Poincaré had realized that some starting conditions can give rise to unusual trajectories and laid a solid mathematical foundation for the study of chaos.

The major drawback to numerical studies in Poincaré's time was the lack of an efficient means of obtaining solutions to the equations of motion. The advent of digital computers with ever-increasing performances has added an experimental approach to the whole study of nonlinear dynamics in general, and solar system dynamics in particular. Combined with new observations and advances in theory, this means that we can now recognize the important role that chaos has played in determining the dynamical structure and evolution of the solar system.

There is still no universally accepted definition of chaos, even though it can be detected in a variety of dynamical systems. However, for our purposes we can make use of the following definition: An object in the solar system can be said to exhibit chaotic motion if its final dynamical state is sensitively dependent on its initial state. Given that the measure of any physical quantity has a built-in error, the lack of precision in starting conditions is transformed into an uncertainty in final conditions. In this chapter we examine the nature and consequences of chaotic phenomena in the context of long-term dynamical evolution. The article by Murray (1998) provides a summary of the major results in this field.

9.2 Sensitive Dependence on Initial Conditions

Consider the motion of a test particle in the vicinity of a planet orbiting the Sun. If the planet is in a circular orbit then this is an example of the circular restricted three-body problem. In the planar case the components of the particle's osculating position and velocity vectors at any time define four quantities, x, y, $\dot{x}$, and $\dot{y}$, which give the particle a unique "position" in a four-dimensional space referred to as *phase space*. As the particle evolves it follows a trajectory in phase space as well as the usual *configuration space* consisting of its motion in the x–y plane. If the particle experiences only the gravitational field of the Sun then its motion would be entirely predictable. However, the perturbations from the planet cause certain regions of the phase space to become chaotic; the

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orbital evolution of test particles placed in such regions will take place in an unpredictable fashion.

Figure 9.1 illustrates this phenomenon for two nearby starting conditions for the orbit of a test particle perturbed by Jupiter in the planar, circular restricted problem. Both orbits take the test particle close to Jupiter and yet a difference of only 0.3° in initial longitude is sufficient to produce a dramatically different outcome. This is an example of what is called the *butterfly effect.* This was first mentioned in the context of chaotic weather systems, whereby it was suggested that under suitable conditions the flapping of a butterfly's wings in one part of the world could ultimately lead to a hurricane elsewhere in the world. In the context of solar system dynamics the test particle can be thought of as a spacecraft receiving a gravity assist from a Jupiter flyby. In such examples of chaos a small change in starting values changes the geometry of the encounter and hence the size of the direct perturbation received from the planet.

Comets and asteroids with sufficiently large eccentricities can intersect planetary orbits. With a semi-major axis $a = 13.75$ AU and an eccentricity $e = 0.385$, asteroid (2060) Chiron (see entry in Table A.17) has a perihelion distance inside the orbit of Saturn and an aphelion distance close to the orbit of Uranus. Using the best available orbital elements Oikawa & Everhart (1979) carried out several numerical integrations of Chiron's orbit, all based on starting conditions close to the accepted values. These showed that Chiron will undergo several close approaches to planets. Based on the fact that different starting conditions showed different outcomes, a characteristic of chaotic motion, Oikawa & Everhart were only able to take a probabilistic approach to determining the ultimate fate of

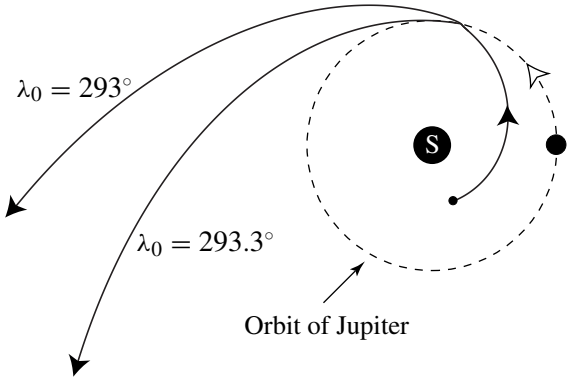


Fig. 9.1. The trajectories of two test particles started with the same semi-major axis ($a_0 = 0.8$), eccentricity ($e_0 = 0.4$), and longitude of pericentre ($\varpi = 295^\circ$) but with slightly different initial mean longitudes ($\lambda_0 = 293^\circ$ and $\lambda_0 = 293.3^\circ$). Jupiter's orbit is denoted by a dashed circle and it has zero initial longitude. The numerical integration lasted one Jupiter period.

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Chiron. They estimated a 1 in 8 chance that Saturn will place Chiron on to a hyperbolic orbit that ejects it from the solar system. The more likely outcome (a 7 in 8 chance) is that the close approaches to Saturn will cause Chiron's orbit to evolve towards the inner solar system where it would come under gravitational perturbations from Jupiter. The classification of Chiron as an asteroid is now debatable with the discovery of large, nonperiodic changes in its brightness (Tholen et al. 1988) and a coma (Meech & Belton 1989). Chiron is now classified as a *Centaur,* one of a number of giant, planet-crossing, cometlike objects that may represent Edgeworth–Kuiper belt objects on their way to becoming short-period comets (see, for example, Luu 1994).

Another example of the butterfly effect in the context of solar system dynamics is the orbital history of Comet Shoemaker-Levy 9. Numerical integrations of the comet's orbit show that prior to its impact with Jupiter in July 1994, it was disrupted by a close approach to Jupiter in 1992. Work by Chodas & Yeomans (1996) suggests that the comet was originally captured into orbit around Jupiter in 1929 $\pm$ 9 y. Prior to capture its orbit was probably similar to that of many of the Jupiter family of comets, with a low-eccentricity orbit that was likely to have been interior to Jupiter's orbit. Further information about the history of the comet is impossible, because (i) the orbit is chaotic and (ii) there will be no new astrometric data to enable an improved initial orbit to be determined.

However, chaos can be more subtle. Figure 9.2 shows the results of two numerical integrations of the equations of motion in the circular restricted three-body problem. In this case the orbital evolution is almost exactly the same for approximately 150 Jupiter periods, after which they gradually differ. There is no obvious reason for the drift (for example, there are no close approaches to Jupiter during the integration period) but the behaviour is again characteristic of chaotic motion with the orbits showing a sensitive dependence on initial conditions.

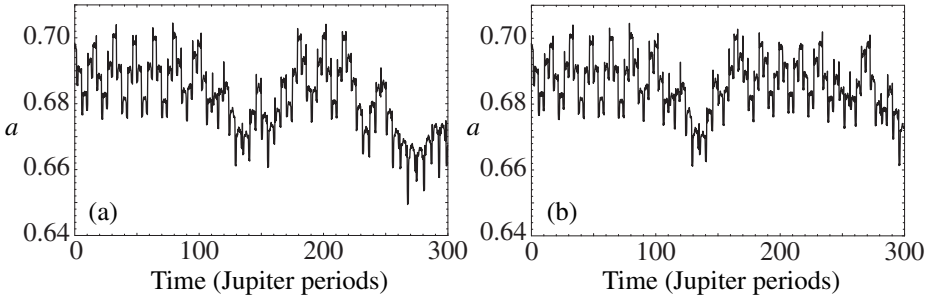


Fig. 9.2. The variation in semi-major axis of two test particles perturbed by Jupiter ($\mu_2 = 0.001$) in the planar, circular restricted problem. The particles were started with the same semi-major axis ($a_0 = 0.6984$), eccentricity ($e_0 = 0.1967$), and longitude of pericentre ($\varpi = 0^\circ$) but with slightly different initial mean longitudes ((a) $\lambda_0 = 0^\circ$ and (b) $\lambda_0 = 10^{-6^\circ}$). Jupiter was started with zero initial longitude.

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The type of chaotic behaviour contrasts with the more dramatic example shown above (see Fig. 9.1) and yet both are valid illustrations of chaos in the circular restricted problem.

9.3 Regular and Chaotic Orbits

We have already seen that the two-body problem is integrable (see Chapter 2) whereas the three-body problem is not (see Chapter 3). In fact there are chaotic solutions to the restricted three-body problem for certain starting conditions but we have deliberately avoided discussing them up to this point. We shall carry out a more thorough examination of chaos in the context of the circular, restricted problem in Sect. 9.4. Before proceeding with these and other examples of chaotic motion in the solar system, we demonstrate the differences between regular (i.e., nonchaotic) motion and chaotic motion, showing the characteristics of each and how some of the properties of chaotic orbits can be quantified.

9.3.1 The Poincaré Surface of Section

We met the concept of a surface of section in Sect. 5.4 when we discussed spin–orbit resonance. In that case we were representing the solution of a second-order, nonlinear differential equation for the time variation of θ, the orientation angle of a nonspherical satellite. In Fig. 5.17b and 5.18b we plotted points denoting the values of $\dot{\theta}/n$ and θ at every pericentre passage, where n was the mean motion of the satellite.

In the case of the planar, circular, restricted three-body problem the situation is more complicated. The equations of motion given in Eqs. (3.16) and (3.17) consist of two, simultaneous, nonlinear, second-order differential equations. The equations are repeated here for convenience:

$$\ddot{x} - 2n\dot{y} - n^2 x = -\mu_1 \frac{x + \mu_2}{r_1^3} + \mu_2 \frac{x - \mu_1}{r_2^3}, \tag{9.1}$$

$$\ddot{y} + 2n\dot{x} - n^2 y = -\left(\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3}\right) y, \tag{9.2}$$

where $\mu_1 = m_1/(m_1 + m_2)$, $\mu_2 = m_2/(m_1 + m_2)$, and

$$r_1^2 = (x + \mu_2)^2 + y^2, \tag{9.3}$$

$$r_2^2 = (x - \mu_1)^2 + y^2, \tag{9.4}$$

where r_1 and r_2 are the distances of the test particle from the masses m_1 and m_2 respectively.

The solution consists of sets of values of x, y, $\dot{x}$, and $\dot{y}$ at a sequence of times, where these quantities denote the components of the position and velocity vectors in the rotating reference frame. Recall that apart from special starting conditions the restricted problem is nonintegrable and we have to resort to numerical solutions of the equations of motion.

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We showed in Chapter 3 that there exists a constant of the motion, the Jacobi constant, defined by

$$C_{\mathrm{J}} = n^2\left(x^2 + y^2\right) + 2\left(\frac{\mu_1}{r_1} + \frac{\mu_2}{r_2}\right) - \dot{x}^2 - \dot{y}^2. \tag{9.5}$$

We have already discussed how the values of x, y, $\dot{x}$, and $\dot{y}$ at any given time correspond to a single point in a four-dimensional phase space. Because of the existence of the Jacobi constant in the circular restricted problem the path of the particle in phase space is confined to a surface (see Fig. 9.3a). Therefore, for a given, fixed value of the Jacobi constant we only require three out of the four quantities to define the osculating orbit at that time uniquely. This is because the other quantity can be determined, at least up to a sign change, by the equation defining the Jacobi constant. For example, if we take x, y, and $\dot{x}$ as our three quantities, the other one, $\dot{y}$, is determined by Eq. (9.5) (see Fig. 9.3b), provided we have noted the value of the Jacobi constant at time $t = 0$. If we now define a plane, say $y = 0$, in the resulting three-dimensional space, the values of x and $\dot{x}$ can be plotted every time the particle has $y = 0$ (see Fig. 9.3c). The ambiguity in the sign of $\dot{y}$ is removed by considering only those crossings with a fixed sign of $\dot{y}$. This is the method of the *Poincaré surface of section* or the *Poincaré map* and this is the technique we have used to illustrate the regular and chaotic regions in the circular restricted problem. The section is obtained by fixing a plane in the phase space and plotting the points when the trajectory intersects this plane in a particular direction. Note that as a result we do not plot the points at equal time intervals; we only plot a point when an intersection takes place.

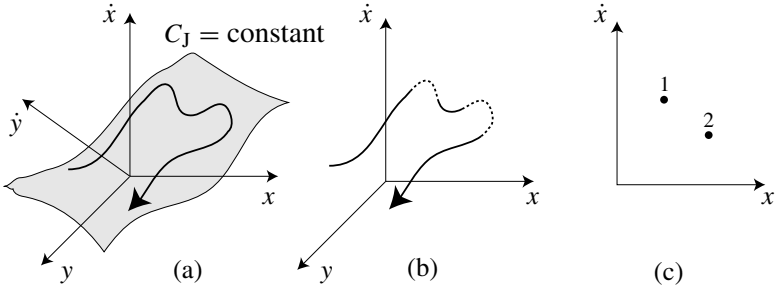


Fig. 9.3. The production of a Poincaré surface of section. (a) The position and velocity of the particle define a point in a four-dimensional phase space at any particular time. The existence of the Jacobi constant means that the test particle's path is moving on a particular surface in this space. (b) We can consider just the x, y, and $\dot{x}$ values, since the value of $\dot{y}$ is always determined up to an arbitrary sign change by Eq. (9.5). The dashed line denotes the part of the trajectory where $y < 0$. (c) A point is plotted every time $y = 0$ with $\dot{y} > 0$, resulting in a sequence of points in a two-dimensional space.

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At first glance we seem to have removed, or at least hidden, a lot of information about the orbit. However, the main advantage of the surface of section method is this very sparseness. Instead of plotting the trajectory in x–y space and following the test particle around numerous orbits, which may only change slightly, with the surface of section method we get an overview of changes in the orbit. All the information is still contained within those points, even though some effort may be required to extract it. There are other advantages to the method.

We are able to use this technique because we are considering the circular, planar, restricted problem. If we now assume that the eccentricity of the secondary mass is nonzero then we can no longer make use of a Poincaré surface of section for the problem. However, we can make use of the averaging principle and isolate only those terms in an expansion of the disturbing function that are likely to dominate. In some circumstances it is possible to obtain analytical solutions of the averaged system and this was one of the techniques employed in our study of resonant perturbations in the previous chapter.

Now we consider numerical solutions to our system of equations using $\mu_2 = 10^{-3}$, a value comparable to the Jupiter/Sun mass ratio.

9.3.2 Regular Orbits

Consider the trajectory of a test particle with starting values $x_0 = 0.55$, $y_0 = 0$, and $\dot{x}_0 = 0$, with $\dot{y} > 0$ determined from the solution of Eq. (9.5) with $C_J = 3.07$. We can convert these values to their equivalent semi-major axis a and eccentricity e using the formulae given in Chapter 2. This gives $a_0 = 0.6944$ and $e_0 = 0.2065$ for the initial values. In the Sun–Jupiter system this value of a_0 would correspond to a test particle at 3.612 AU, well beyond the main belt of asteroids.

In Fig. 9.4a we show the evolution of e as a function of time. Although there are obvious variations, the plot shows a regular variation in the eccentricity in the

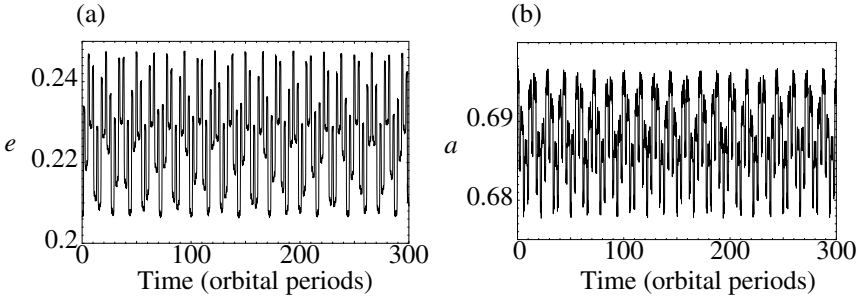


Fig. 9.4. The time variability of (a) eccentricity e and (b) semi-major axis a for initial values $a_0 = 0.6944$ and $e_0 = 0.2065$. The plots show a behaviour characteristic of regular orbits. (Adapted from Murray 1998.)

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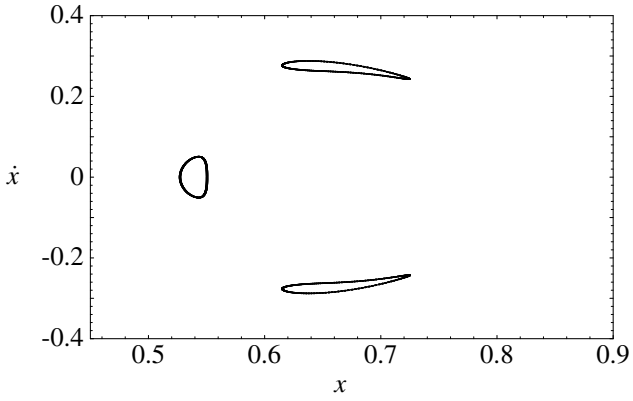


Fig. 9.5. A surface of section for the regular trajectory shown in Fig. 9.4. The points are plotted whenever $y = 0$ with $\dot{y} > 0$. (Adapted from Murray 1998.)

range 0.206–0.248 over the course of the integration. At this particular location we would expect to see resonant perturbations since $(a/a_J)^{3/2} = 0.564 \approx 4/7$ and the orbit of the test particle is close to a 7:4 resonance with Jupiter. The equivalent plot for the variation of the semi-major axis is shown in Fig. 9.4b. Note that the changes in a are correlated with those in e, as we would expect from some of the simple relations discussed in Sect. 8.5. The value of a appears to be librating close to the location of the nominal resonance at $a = (4/7)^{2/3} \approx 0.689$. Figure 9.4 only shows the variations in e and a. A complete description of the trajectory should also involve plots of ϖ and another angle, such as the mean longitude λ.

Figure 9.5 shows the same trajectory displayed as a surface of section, obtained by plotting the values of x and $\dot{x}$ whenever $y = 0$ with $\dot{y} > 0$. The pattern that emerges shows three, distinct "islands". The appearance of such islands is one characteristic of resonant motion when displayed as a surface of section. In these cases a mean motion resonance of the form $p + q : p$, where p and q are integers, produces q islands. In this case $p = 4$, $q = 3$, and three islands are visible. An important point to note is the manner in which these islands appear on the plot as the trajectory is followed. Rather than tracing out one island at a time, successive points occur at each of the three island locations in turn, until they appear to form three smooth curves.

If we choose a value of x_0 that places the starting point at the centre of the island that straddles the $\dot{x} = 0$ line, then the trajectory would appear as a succession of three points, one at the centre of each island in turn. This is because the centre of each island corresponds to a starting condition that places the test particle at the middle of the resonance. This is the type of resonant orbit illustrated in Fig. 8.15a for the 2:1 resonance. Such points are said to be periodic points of the Poincaré map because the system returns to the same point every third time

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the trajectory crosses the section. By moving the starting location further away from the centre the islands would get larger, corresponding to larger variations in e and a. Eventually some starting values would lead to trajectories that were not in resonant motion and there would no longer be distinct islands in the section plot.

9.3.3 Chaotic Orbits

Figure 9.6 shows the plots of e and a as a function of time for the test particle orbit with starting values $x_0 = 0.56$, $y_0 = 0$, and $\dot{x}_0 = 0$, and $\dot{y}$ determined from Eq. (9.5) with $C_J = 3.07$, the same value used above to illustrate the regular trajectory. The corresponding orbital elements are $a_0 = 0.6984$ and $e_0 = 0.1967$. Note that although these values are only slightly different from those used above, the nature of the variations in e and a are very different. The eccentricity undergoes irregular variations from 0.188 to 0.328 and the value of a does not remain close to the resonant value. This is a chaotic trajectory where there is no obvious pattern to the variations in the orbital elements; this alone does not necessarily mean that the orbit is chaotic. For example, a secular solution to the N-body problem can display complicated behaviour similar to that associated with chaotic motion and yet we know that it is derived from an analytical solution composed of a finite number of frequencies.

The identification of this particular orbit as chaotic becomes apparent from a study of its Poincaré surface of section (see Fig. 9.7). Note that the orbit covers a larger region of phase space than in the regular example (cf. Fig: 9.5). Furthermore, instead of the points lying on a smooth curve, they are beginning to fill an area of the phase space. Note also that there is a tendency for some points to "stick" to the edge of the 7:4 and other resonances; indeed the points

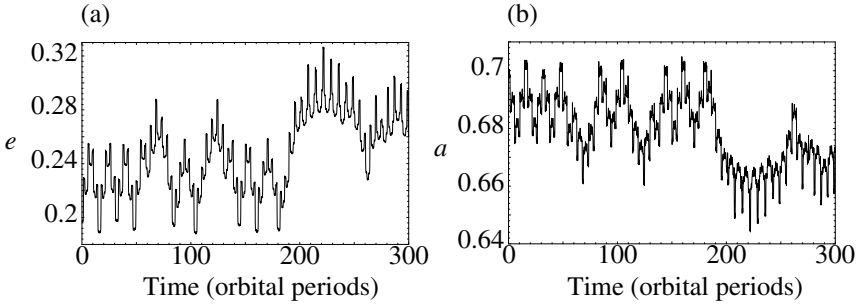


Fig. 9.6. The time variability of (a) eccentricity e and (b) semi-major axis a for initial values $a_0 = 0.6984$ and $e_0 = 0.1967$. The plots show a behaviour characteristic of chaotic orbits. (Adapted from Murray 1998.)

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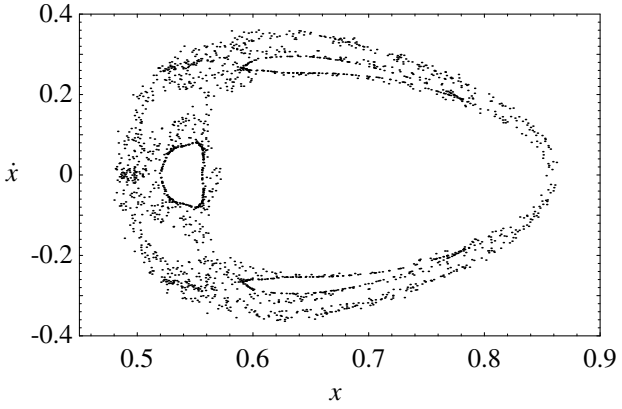


Fig. 9.7. A surface of section for the chaotic trajectory shown in Fig. 9.6. The points are plotted whenever $y = 0$ with $\dot{y} > 0$, and the scale is the same as that in Fig. 9.5. (Adapted from Murray 1998.)

in the chaotic trajectory help to define several empty regions, each of which can be associated with a resonance. The stickiness phenomenon implies that if we examine the trajectory at particular times it may give the impression of regular motion. In fact, this is another characteristic of chaotic behaviour – it can give the impression of regular motion for long periods of time; this can make chaotic motion difficult to detect in some circumstances.

Although the chaotic orbit shown in Fig. 9.7 covers a large area of the x–$\dot{x}$ space, there is some evidence that there are still bounds to the motion. Thus, this orbit may illustrate the phenomenon of *bounded chaos* and this will be discussed in Sect. 9.4.

9.3.4 The Lyapounov Characteristic Exponent

Chaotic orbits have the characteristic that they are sensitively dependent on initial conditions. This is illustrated in Fig. 9.8 where we show part of the plot of e as a function of time for two similar chaotic trajectories. The first corresponds to the trajectory shown in Figs. 9.6 and 9.7 where we used $x_0 = 0.56$; the second has $x_0 = 0.56001$ and we have adjusted the initial value of $\dot{y}$ to maintain the same value of C_J. It is clear that after 60 orbital periods the orbits have drifted apart. If we performed the same displacement experiment and followed the evolution of a regular orbit there would be some variation in the magnitude of the relative separation but we would not detect such a large drift.

We can make use of the type of divergence seen in Fig. 9.8 to measure the *maximum Lyapounov characteristic exponent* (LCE) of a dynamical system, giving us a quantitative measure of the rate of divergence of nearby trajectories (see, for example, Lichtenberg & Lieberman 1983). In a dynamical system such

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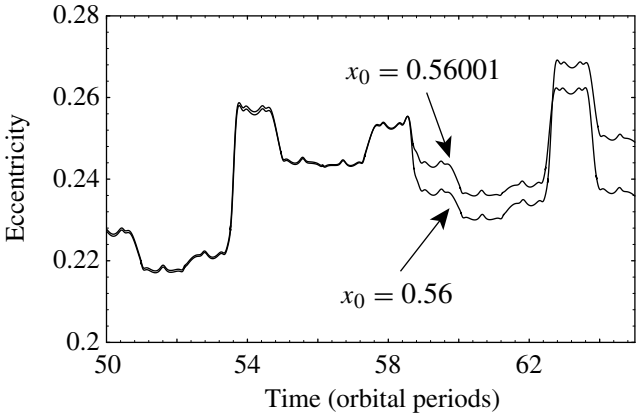


Fig. 9.8. Parts of two plots of the time evolution of the eccentricity for two nearby orbits in a chaotic region of the phase space. The initial values of x differ by 0.00001 and a clear separation can be seen after ~ 60 orbital periods. (Adapted from Murray 1998.)

as the three-body problem there are a number of quantities called the Lyapounov characteristic exponents. For an arbitrary starting condition it can be shown that a measurement of the local divergence of nearby trajectories leads to an estimate of the largest of these exponents.

Consider two orbits separated in phase space by a distance d_0 at time t_0 (see Fig. 9.9a). Let d be their separation at time t. The orbit is chaotic if d is approximately related to d_0 by

$$d = d_0 \exp \gamma (t - t_0), \tag{9.6}$$

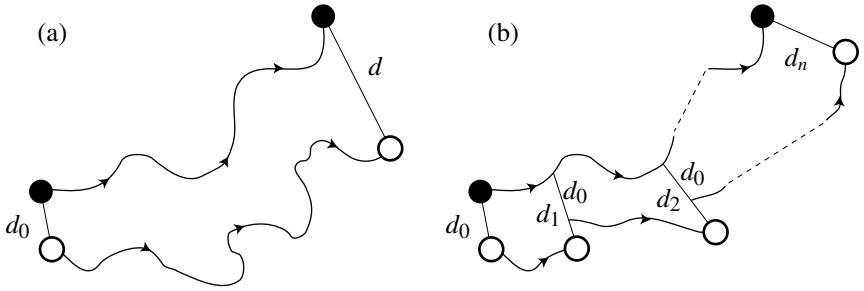


Fig. 9.9. Calculation of the maximum Lyapounov characteristic exponent by measuring the divergence of nearby trajectories. (a) The straightforward method of calculating the initial and final separations in phase space. (b) The renormalisation method whereby the displaced particle's trajectory is moved back along the separation vector to the original displacement distance.

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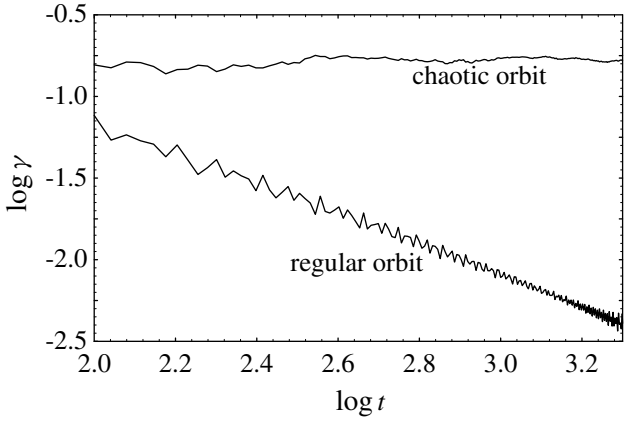


Fig. 9.10. Plots of $\log \gamma$ as a function of log time for the regular and chaotic trajectories discussed above. (Adapted from Murray 1998.)

where γ is the maximum Lyapounov characteristic exponent. Note that we must have $\gamma > 0$ for otherwise the trajectories would approach one another as t increased. We can estimate the value of γ from the results of a numerical integration by means of the relation

$$\gamma = \lim_{t \to \infty} \frac{\ln (d/d_0)}{t - t_0}. \tag{9.7}$$

Monitoring the behaviour of γ as a function of time on a log–log scale usually reveals a striking difference between regular and chaotic trajectories. For regular orbits the initial and final displacements are close to one another ($d \approx d_0$) and hence a log–log plot would have a slope of -1. However, if the orbit is chaotic, then γ tends to a positive value. In cases where the displaced trajectory drifts too far from the original one γ is no longer a measure of the local divergence of the orbits. Therefore it is advisable to rescale (or renormalise) the separation vector of the nearby trajectory at fixed time intervals, Δt (see Fig. 9.9b). If there are n such renormalisations then the revised estimate of γ is

$$\gamma = \lim_{n \to \infty} \sum_{i=1}^{n} \frac{\ln (d_i/d_0)}{n \Delta t}. \tag{9.8}$$

Figure 9.10 shows $\log \gamma$ as a function of $\log t$ for the sample regular and chaotic orbits described above. From this plot we can derive an estimate of $\gamma = 10^{-0.77}$ (orbital periods)$^{-1}$ for the maximum Lyapounov characteristic exponent of the chaotic orbit. The corresponding *Lyapounov time*, the time for the displacement to increase by a factor e, is given by $1/\gamma$, which in this case is ~ 6 orbital periods. Therefore, at least for this starting condition, the chaotic nature of the orbit quickly becomes apparent. This is consistent with the demonstration shown in Fig. 9.8.

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For simpler dynamical systems it can be possible to obtain all the Lyapounov characteristic exponents by analytical means. However, this is usually the exception in solar system dynamics. Furthermore, numerical experiments that follow the evolution of two nearby orbits may not always be successful, since our approximation to the maximum LCE, γ, is only defined in the limit as $t \to \infty$ or $n \to \infty$. For example, the phenomenon of some chaotic orbits sticking to the boundaries of resonances implies that there may be the appearance of a regular trajectory for long periods of time. Therefore, although the chaotic nature of a chaotic orbit may be readily apparent, it may never be possible to prove a regular orbit is truly regular.

9.4 Chaos in the Circular Restricted Problem

The planar, circular, restricted three-body problem is perhaps the simplest dynamical model that approximates the motion of real objects in the solar system and yet, as we have seen, it has orbital solutions that display a surprising degree of complexity. In the previous section we demonstrated the existence of regular and chaotic orbits for a small sample of orbits; yet the structure of the phase space in the circular restricted problem is far from being fully understood. In this section we consider some of the properties of the restricted problem for the specific mass ratio of $\mu_2 = 10^{-3}$, following the work of Winter & Murray (1994a,b). Note that for this value of μ_2 the results given here for the motion of test particles approximate the motion of asteroids perturbed by a coplanar Jupiter moving on a circular orbit.

There have been several numerical studies of the circular restricted problem (see, for example, Hénon 1969 and Jefferys 1971). In the comprehensive survey by Winter & Murray (1994a,b) several hundred surfaces of section were produced for a variety of Jacobi constants and orbits both internal and external to the orbit of the perturber were examined. From these plots it is possible to measure properties such as the location and size of regular and chaotic regions and the maximum amplitude of resonant librations.

Recall that in the restricted problem neither the orbital energy nor the angular momentum are conserved because the potential experienced by the particle is explicitly time dependent. However, the dynamical system retains an integral of the motion, the Jacobi constant, given by Eq. (9.5). As we saw in Chapter 3, setting $\dot{x} = \dot{y} = 0$ in the definition of C_J results in an equation that describes the zero-velocity curves associated with the value of C_J. These curves define the limits of prohibited regions in the physical space since particles with that value of C_J cannot move inside such regions. Figure 9.11 shows zero-velocity curves and the corresponding regions from which the particle is excluded for a selection of values of C_J.

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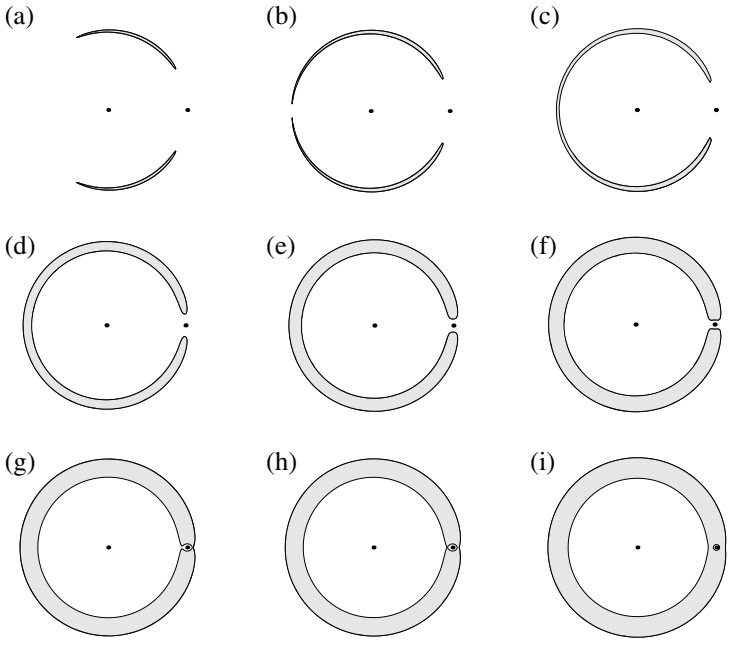


Fig. 9.11. Plots of zero-velocity curves and the enclosed prohibited regions (shaded) values of the Jacobi constant equal to (a) 3, (b) 3.001, (c) 3.002, (d) 3.01, (e) 3.02, (f) 3.03, (g) 3.039, (h) 3.04, and (i) 3.05. The small points denote the locations of the two masses.

For low values of C_J the prohibited regions are confined to a zone around the L_4 and L_5 Lagrangian points located 60° ahead of and behind the orbit of the perturbing mass. For larger values of C_J these zones merge and encompass the L_3 point. However, there is still an open region close to the location of the perturbing mass. As C_J increases the prohibited region broadens and encroaches upon the perturbing mass, eventually enclosing it. At this stage particle orbits that are initially interior to the perturbing mass must remain interior to it for all time; similarly those orbits that are initially exterior to it must remain exterior for all time. For some values of the Jacobi constant (e.g., $C_J = 3.050$) there is a small region of possible motion around the perturbing mass. In such cases a test particle orbiting the perturbing mass can never orbit the central mass or escape from its orbit around the perturbing mass. For larger values of C_J this region decreases further and the radial extent of the forbidden region grows.

It is clear from Fig. 9.11 that for values of $C_J \geq 3.040$ a particle that is started in an orbit about the central mass cannot get closer to the perturber than a certain limit, which is determined by the extent of the prohibited region.

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Figure 9.12 shows the minimum possible distance, $d_{\min}$, between the particle and the perturbing mass as a function of the Jacobi constant, C_J.

In the computations of Winter & Murray the starting conditions were mostly chosen such that for each value of C_J the values of x were selected by taking $y = \dot{x} = 0$ with $\dot{y} > 0$ and Jupiter along the x axis at conjunction with the particle. Hence the integration was usually started at either the pericentre or apocentre of the particle's orbit. In a few cases the starting conditions were chosen such that $\dot{x} \neq 0$. This was needed in order to obtain surfaces of section of regular areas related to the main resonances of even order.

There is always a choice of surfaces of section. In their study Winter & Murray chose a system where the values of x and $\dot{x}$ were computed whenever the trajectory crossed the plane $y = 0$ with $\dot{y} > 0$. In the previous section we saw how the surface of section technique is good at illustrating the regular or chaotic nature of the trajectory. If there are smooth, well-defined islands, then the trajectory is likely to be regular and the islands correspond to libration around an exact resonance between the mean motions of the perturber and the particle; in general the number of islands corresponds to the order of the resonance. Exceptions to this rule include those orbits associated with near-zero eccentricity of the test particle and a class of eccentric orbits in the 1:1 resonance with the perturber. Any "fuzzy" distribution of points in the surface of section implies that the trajectory is chaotic. The surface of section is also useful in demonstrating the extent of the particle's motion throughout the phase space. Throughout this section we show a representative sample of orbits and the appearance of their Poincaré surfaces of section. In the surfaces of section periodic orbits give rise to fixed points that are the centre of islands of stability. The islands correspond to the quasi-periodic orbits librating around the stable positions.

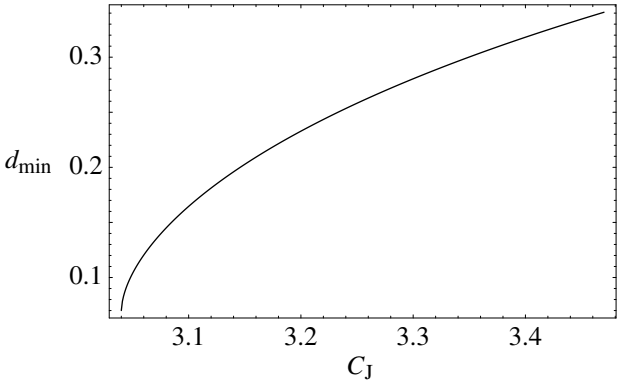


Fig. 9.12. The minimum possible distance $d_{\min}$ between the test particle and the perturbing mass as a function of the Jacobi constant C_J. (Adapted from Winter & Murray 1994a.)

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For a given point, $(x, \dot{x})$, on the surface of section one can use the value of the Jacobi constant, the fact that $y = 0$, and Eq. (9.5) to find the value of $\dot{y}$. Therefore, all four quantities are known and the semi-major axis and the eccentricity can be found using

$$a = \left[\frac{2}{r_1} - \frac{V^2}{\mu_1}\right]^{-1}, \tag{9.9}$$

$$e = \sqrt{1 - \frac{h^2}{\mu_1 a}}, \tag{9.10}$$

where $V^2 = \dot{x}^2 + (\dot{y} + x + \mu_2)^2$ and $h = (x + \mu_2)(\dot{y} + x + \mu_2)$. Using these equations one can identify the initial mean motion frequencies associated with points on a surface of section and then identify the resonances associated with the islands of stability that appear in the plots, bearing in mind that the mean motion of the perturber is equal to unity and that generally the number of islands is equal to the order of the resonance.

The periodic orbits associated with the planar, circular, restricted three-body problem can be classified into two types. *Periodic orbits of the first kind* arise from the fact that the test particle is started in a circular orbit ($e_0 = 0$). Figure 9.13 shows the values of C_J as a function of x_0 for starting conditions with $\dot{x}_0 = 0$ that give rise to periodic orbits of the first kind. This diagram can be used in conjunction with the surface of section plots to identify islands associated with such periodic orbits. The fact that this function has a minimum at $C_J = 3.027$ implies that for a Jacobi constant below this value there are no starting values x_0 that give rise to periodic orbits of the first kind. *Periodic orbits of the second*

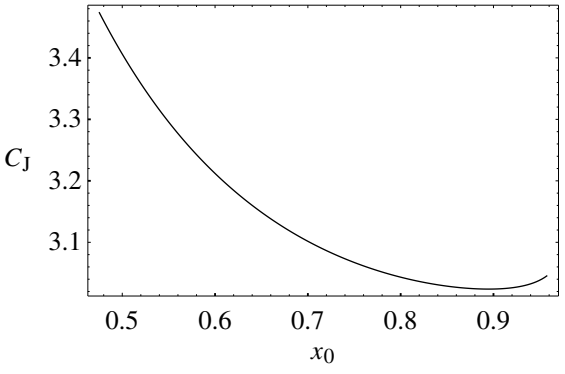


Fig. 9.13. The relationship between the Jacobi constant C_J and the starting value x_0 with $\dot{x}_0 = 0$, which give rise to periodic orbits of the first kind. (Adapted from Winter & Murray 1994a.)

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kind arise from the fact that the particle is at the centre of a resonance; this is true for both primary and secondary resonances (see below).

The structure of the phase space of the planar, circular, restricted three-body problem is mainly determined by resonant phenomena (see Chapter 8). In these situations there is a commensurability between the mean motion of the particle and the mean motion of the perturbing mass, that is a relationship of the form $(p+q)n' \approx pn$, where p and q are integers and n and n' are the mean motions of the test particle and the perturber, respectively; q is called the *order* of the resonance. The location of stable and unstable equilibrium points due to each resonance, as well as the strength of the resonances, define the location and size of regular and chaotic regions. The resonances can be classified into *primary* and *secondary resonances*. Primary resonances are due to commensurabilities between the mean motions of the perturbing mass and the test particle. In the restricted problem secondary resonances arise when the libration frequency of a primary resonance is commensurate with the circulation frequency of a higher order primary resonance.

A set of three islands are shown in Fig. 9.14a corresponding to a single trajectory with $C_J = 3.02$, $x_0 = 0.205$, and $\dot{x}_0 = 0$; this corresponds to $a_0 = 0.521$ and $e_0 = 0.606$. This trajectory is librating around the 5:2 primary resonance. Clusters of islands arising from a simpler island chain are features that appear in surfaces of section due to a trajectory librating around a secondary resonance. Figure 9.14b shows the surface of section of a trajectory that librates around the

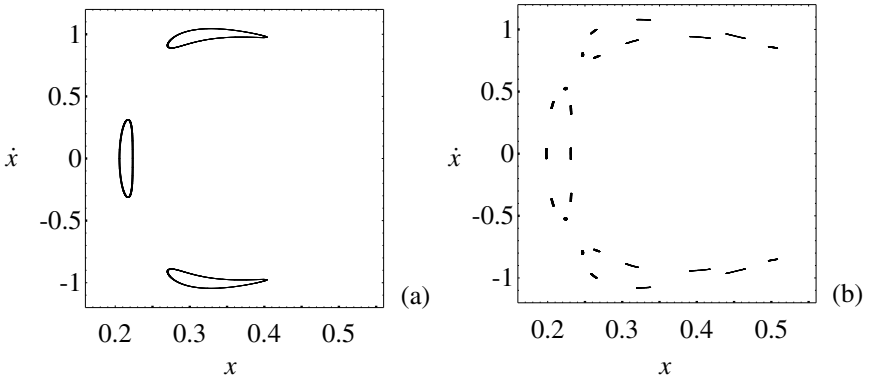


Fig. 9.14. Poincaré surfaces of section showing two quasi-periodic orbits for the same value of the Jacobi constant, $C_J = 3.02$. (a) Using $x_0 = 0.205$ the trajectory shows three islands associated with libration about the 5:2 primary resonance. (b) Using $x_0 = 0.23$ the trajectory shows the twenty-four islands associated with the 8:1 secondary resonance between the libration frequency of the 5:2 primary resonance and the circulation frequency of the 60:24 primary resonance. (Adapted from Winter & Murray 1994a.)

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8:1 secondary resonance arising from libration in the 5:2 primary resonance and circulation in the 60:24 primary resonance.

The chaotic motion is associated with unstable equilibrium points and their accompanying separatrices. The existence of chaotic motion does not imply that the trajectory is unbounded. For example, Fig. 9.11 shows that orbits can remain confined for all time for certain values of the Jacobi constant. Chaotic trajectories can be localised or cover large regions of the $(x, \dot{x})$ plane. Two extreme examples are shown in Figure 9.15. The first shows the motion of a particle close to a narrow separatrix (Fig. 9.15a); the second case (Fig. 9.15b) shows a chaotic "sea" with the particle covering a large area. In fact chaotic trajectories can be used to highlight the locations of regular regions of the phase space.

An inspection of the surfaces of section on the following plots shows the clear increase in the extent of the chaos as C_J decreases. This is connected with the change in the zero-velocity curves (see Fig. 9.11) that permits closer approaches between the particle and the perturber as the value of the Jacobi constant gets smaller. From Eq. (9.5) it is clear that if $(x, \dot{x})$ denotes a point on the surface of section, then the point $(x, -\dot{x})$ must also lie on the same surface. This explains the obvious symmetry of the surfaces of section about the x axis.

In general, for the starting conditions considered by Winter & Murray (1994a,b), the orbits are initially prograde in the inertial and rotating reference frames. Most prograde orbits in the rotating frame only produce points on the surface of section with positive values of x. However, points with negative values of x can also appear in a surface of section due to regular orbits with apocentric

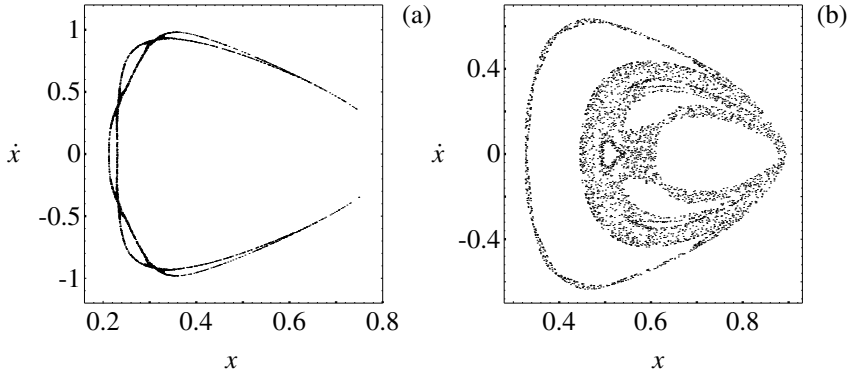


Fig. 9.15. Poincaré surfaces of section for two chaotic orbits. (a) With $C_J = 3.111$ and $x_0 = 0.23$ the resulting trajectory is at the separatrix of the 8:3 resonance and remains confined to a well-defined area. (b) With $C_J = 3.051$ and $x_0 = 0.445$ the plot shows the large area covered in the $(x, \dot{x})$ space by a chaotic trajectory; notice the regular regions are being highlighted by the chaotic trajectory. (Adapted from Winter & Murray 1994a.)

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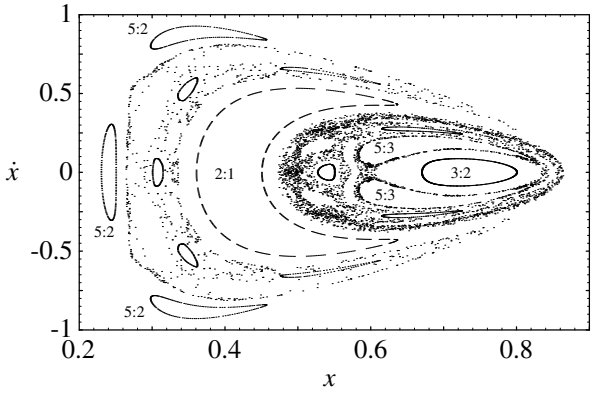


Fig. 9.16. Poincaré surface of section for a variety of trajectories with $C_J = 3.07$. The location of several resonant islands are identified. (Adapted from Murray 1998.)

distances larger than unity. Chaotic trajectories can achieve mean motions that are lower than that of the perturber or even orbits that are retrograde in the inertial frame. Both cases give rise to orbits that are retrograde in the rotating frame, producing points on the surface of section with negative values of x. These points have been deliberately excluded from the plots presented here.

The extent of the chaos in a system can depend on several factors. In the case of the circular restricted problem the key quantities are the values of C_J and μ_2. In Figs. 9.16 and 9.17 a number of trajectories are shown for two different values of the Jacobi constant but the same value of μ_2. In the first case, Fig. 9.16, the value is $C_J = 3.07$ (the same as the value used in Figs. 9.5 and 9.7) while in Fig. 9.17 it is $C_J = 3.13$. It is clear that the extent of the chaos

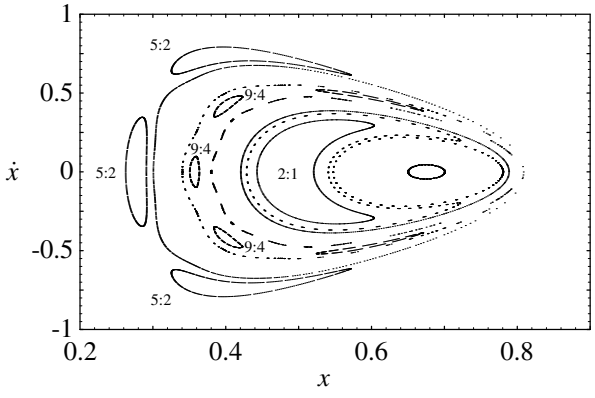


Fig. 9.17. Poincaré surface of section for a variety of trajectories with $C_J = 3.13$. The location of several resonant islands are identified. (Adapted from Murray 1998.)

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is reduced in Fig. 9.17. We have already seen in Fig. 9.11 that the value of C_J determines how close the test particle can get to the perturber. Indeed, for the case $\mu_2 = 0.001$ and $C_J > 3.04$ it is impossible for their orbits to intersect, although the perturbations can still be significant. We can interpret the behaviour seen in Figs. 9.16 and 9.17 in the following terms: As C_J decreases, the extent of the chaotic separatrices associated with each resonance increases until the regular curves separating adjacent resonances are broken down and the neighbouring chaotic regions begin to merge. This can also be thought of as the overlap of adjacent resonances giving rise to chaotic motion and will be discussed further in Sect. 9.6. It is the overlapping that permits chaotic orbits to explore regions of the phase space that are inaccessible to the regular orbits. If we apply this observation to the Sun–Jupiter–asteroid problem, it implies that certain asteroids might undergo large changes in their orbital elements.

9.5 Algebraic Mappings

We have seen how trajectories of the test particle in systems such as the restricted three-body problem are derived by finding numerical solutions of the equations of motion. This allows us to calculate the position of the particle in phase space at some time t from knowledge of its initial state at some time t_0. The numerical method to accomplish this (e.g., the standard fourth-order Runge–Kutta algorithm) consists of a sequence of algebraic instructions that take x_0, y_0, $\dot{x}_0$, $\dot{y}_0$ and return x, y, $\dot{x}$, and, $\dot{y}$. Any such numerical method is an example of an *algebraic mapping,* whereby the evolution of the system can be followed by means of a mapping from the initial state to the final state.

In this section we examine several algebraic mappings that have played a key role in helping us understand dynamical evolution in the solar system. Here we are not concerned with new numerical methods as such; these mappings represent tools that allow us to study orbital evolution over much longer timescales than permitted by traditional numerical methods. However, in common with all such methods they only represent an approximation to the motion. Because chaotic motion can sometimes only be revealed after long integrations, the advent of efficient new methods for studying orbital evolution has played an important role in solar system dynamics.

9.5.1 The Standard Map

First we consider the *standard map* (see Chirikov 1979). This was derived to approximate the motion of charged particles in accelerators. Although it may not seem directly applicable to the solar system, the standard map arises from a model of a perturbed pendulum and we have already noted that motion near resonance can be approximated by a pendulum equation (see Sect. 8.6). In fact, the derivation and dynamical properties of the standard map prove to be a simple yet useful guide to similar mappings that are even more applicable. Indeed, the

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complexity of orbits that result from such a simple system provides a foretaste of what can be expected from other mappings.

Using a suitable scaling of the coordinates we can write the Hamiltonian of the simple pendulum as

$$\mathcal{H} = \frac{I^2}{4\pi} + \frac{k_0}{2\pi}\cos\theta, \tag{9.11}$$

where I and θ are action and angles variables derived from the original angular position and momentum (effectively the φ and $\dot{\varphi}$ in our original pendulum model given in Eq. (8.48)). The details of the appropriate transformations need not concern us here since we are only interested in the properties of the system. Now suppose that the suspension point of the pendulum is subjected to an oscillation that can be represented by an infinite number of high-frequency terms such that the Hamiltonian is given by

$$\mathcal{H} = \frac{I^2}{4\pi} + \frac{k_0}{2\pi}\cos\theta + \sum_{n=1}^{\infty} k_n(I)\cos(\theta - nt). \tag{9.12}$$

This makes the problem difficult to handle since we have to deal with an infinite number of short-period terms. However, with suitable approximations we can make use of these infinite number of terms to introduce an impulse into the system. Effectively we will be considering a new system where the cosine term is only applied at discrete intervals. The resulting system can be approximated by an algebraic map. If we write

$$\frac{k_0}{2\pi}\cos\theta + \sum_{n=1}^{\infty} k_n(I)\cos(\theta - nt) \approx \frac{k_0}{2\pi}\sum_{n=0}^{\infty}\cos(\theta - nt) \tag{9.13}$$

then we can use the properties of the Dirac δ-function to obtain

$$\mathcal{H} \approx \frac{I^2}{4\pi} + k_0\cos\theta\,\delta_{2\pi}(t), \tag{9.14}$$

where $\delta_{2\pi}(t)$ is a 2π-periodic δ-function.

The equations of motion are then

$$\dot{I} = -\frac{\partial\mathcal{H}}{\partial\theta} = +k_0\sin\theta\,\delta_{2\pi}(t), \tag{9.15}$$

$$\dot{\theta} = +\frac{\partial\mathcal{H}}{\partial I} = \frac{I}{2\pi}. \tag{9.16}$$

Integrating Eq. (9.15) gives

$$I' - I = k_0\sin\theta\int_0^{2\pi}\delta_{2\pi}(t)\,\mathrm{d}t = k_0\sin\theta \tag{9.17}$$

while Eq. (9.16) gives

$$\theta' - \theta = \int_0^{2\pi}\frac{I}{2\pi}\,\mathrm{d}t \approx I'. \tag{9.18}$$

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Thus, with time steps of 2π, we can write this approximate solution of the equations of motion as a mapping taking us from (I, θ) to (I', θ'). This is the *standard map* given by

$$\begin{aligned} I' &= I + k_0 \sin\theta, \\ \theta' &= \theta + I'. \end{aligned} \tag{9.19}$$

The standard map is a discretised approximation to the motion of the perturbed pendulum and defines a completely deterministic system; that is, after any number of iterations we can, in principle, state what the values of I and θ will be.

If we always take I and θ as modulo 2π (thanks to our original scaling) then we can consider the resulting motion of the perturbed pendulum as motion on the surface of a torus, with a different torus for each value of k_0; this is because any point on the torus can be defined by two angles. The parameter k_0 can be thought of as the magnitude of the perturbation on the pendulum.

Figure 9.18 shows thirty-six separate trajectories for two values of the constant k_0. The starting conditions were identical in each case. For both values of k_0 the island at $(0, \pi)$ is present. This is the stable equilibrium point in the original unperturbed pendulum that survives the short-period perturbations. However, even for $k_0 = 0.8$ we can see that a number of the islands have emerged. They seem to occur in chains and as with the circular restricted problem they represent different resonances between the fundamental frequency of the unperturbed pendulum and the infinite number of short-period perturbing frequencies. Again the number of islands in the chain is related to the order of the resonance. At $k_0 = 1.2$ the unusual behaviour of the orbits at the edges of the resonant islands becomes obvious; this is chaotic motion near the separatrix. As k_0 increases it seems as though the adjacent islands are starting to merge as the extent of the "fuzzy" separatrix motion increases. For $k_0 = 1.2$ there are few islands left and

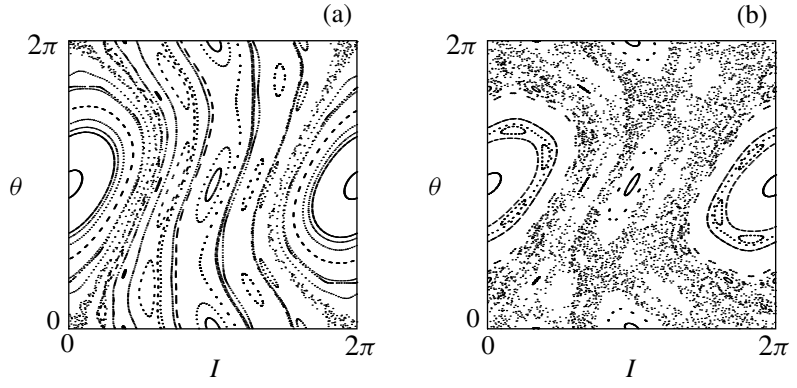


Fig. 9.18. The evolution of thirty-six different trajectories of the standard map with different values of k_0. (a) $k_0 = 0.8$. (b) $k_0 = 1.2$.

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the irregular motion trajectories are starting to dominate, and they appear to have few bounds. Note how the maximum extent of the islands become well defined as the trajectory traces them out for short periods. Most of these phenomena have already been observed in our study of surfaces of section in the circular restricted problem, and yet here they arise from a much simpler system.

9.5.2 Resonance Maps

Resonance maps are derived from the averaged disturbing function in order to model the evolution of a test particle at a given resonance. They rely on the fact that the secular part of the disturbing function can be solved analytically (see Sect. 7.9) while the individual resonant terms can be approximated by a sequence of impulses applied in succession. The technique for converting the resonant terms to impulses follows that used by Chirikov (1979) to derive the standard map.

Wisdom (1982) pioneered this method when he devised an algebraic mapping to study the orbital evolution of test particles at the 3:1 jovian resonance in the asteroid belt. In particular he was interested in explaining the origin of the prominent gap in the asteroid distribution associated with this resonance. In its original form Wisdom's map included variations in the inclination and allowed for secular evolution of Jupiter's orbit using the theory developed by Brouwer & van Woerkom (1950). In a subsequent paper Wisdom (1983) presented a revised version of the map to study the planar case with fixed values of Jupiter's eccentricity and pericentre. Here we show how the planar version of the map was derived; examples of its use in understanding the structure of the 3:1 gap are given in Sect. 9.8.2.

Wisdom's starting point was a second-order expansion of the disturbing function containing the secular and resonant terms. He transformed the standard orbital elements to Poincaré elements given by

$$L = \sqrt{\mu_1 a}, \tag{9.20}$$

$$\rho = \sqrt{\mu_1 a}\left(1 - \sqrt{1-e^2}\right) \approx \sqrt{\mu_1 a}\frac{e^2}{2}, \tag{9.21}$$

where $\mu_1 = 1 - \mu$ and the conjugate elements are λ and ϖ respectively. He then transformed to the resonant variable $\varphi = \lambda - 3\lambda'$ and used $\omega = -\varpi$, $x = \sqrt{2\rho}\cos\omega$, and $y = \sqrt{2\rho}\sin\omega$ to produce the Hamiltonian

$$\begin{aligned}\mathcal{H} = & -\frac{\mu_1^2}{2\Phi^2} - 3\Phi + \Lambda' + F_1(x^2+y^2) + \bar{F}_1 e' x \\ & - \left[C_1(x^2-y^2) + D_1 e' x + E_1 e'^2\right]\cos\varphi \\ & + \left[C_1 2xy + D_1 e' y\right]\sin\varphi,\end{aligned} \tag{9.22}$$

where Λ' is the variable conjugate to λ', and F_1, $\bar{F}_1$, C_1, D_1, and E_1 are constants that are functions of Laplace coefficients. Here e' is the eccentricity of Jupiter,

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which can be treated as constant or can be calculated from a known secular solution for the planets. In a procedure similar to the derivation of the standard map, Wisdom added high-frequency terms to the sine and cosine terms in $\mathcal{H}$. Since an infinite number of high-frequency terms are effectively ignored when producing an averaged disturbing function, the inclusion of a selected number of such high-frequency terms could be justified provided they did not change the nature of the problem. In this case they permitted the derivation of a map to study the motion.

The Hamiltonian becomes

$$\begin{aligned}\mathcal{H} = & -\frac{\mu_1^2}{2\Phi^2} - 3\Phi + \Lambda' + F_1(x^2+y^2) + \bar{F}_1 e' x \\ & - \sum_{i=-\infty}^{\infty} C_1(x^2-y^2)\cos[\varphi - i(t-\zeta_1)] + \sum_{i=-\infty}^{\infty} D_1 e' x \cos[\varphi - i(t-\zeta_2)] \\ & + \sum_{i=-\infty}^{\infty} E_1 e'^2 \cos[\varphi - i(t-\zeta_3)] + \sum_{i=-\infty}^{\infty} C_1 2xy \cos[\varphi - i(t-\gamma_1) - \pi/2] \\ & + \sum_{i=-\infty}^{\infty} D_1 e' y \cos[\varphi - i(t-\gamma_2) - \pi/2],\end{aligned} \tag{9.23}$$

where ζ_1, ζ_2, ζ_3, γ_1, and γ_2 are all arbitrary constants. Each sum can now be expressed as a sum of Dirac δ-functions. For example,

$$\begin{aligned}\sum_{i=-\infty}^{\infty} \cos[\varphi - i(t-\zeta_1)] &= \cos\varphi \sum_{i=-\infty}^{\infty} \cos[i(t-\zeta_1)] \\ &= \cos\varphi \sum_{i=-\infty}^{\infty} 2\pi\delta[(t-\zeta_1) - 2\pi i] \\ &= \cos\varphi\, 2\pi\, \delta_{2\pi}(t-\zeta_1),\end{aligned} \tag{9.24}$$

where, as with the standard map, $\delta_{2\pi}$ is a 2π-periodic δ-function.

The planar averaged Hamiltonian now becomes

$$\begin{aligned}\mathcal{H} = & -\frac{\mu_1^2}{2\Phi^2} - 3\Phi + \Lambda' + F_1(x^2+y^2) + \bar{F}_1 e' x \\ & - C_1(x^2-y^2)\cos\varphi\, 2\pi\, \delta_{2\pi}(t-\zeta_1) \\ & - D_1 e' x \cos\varphi\, 2\pi\, \delta_{2\pi}(t-\zeta_2) - E_1 e'^2 \cos\varphi\, 2\pi\, \delta_{2\pi}(t-\zeta_3) \\ & + C_1 2xy \sin\varphi\, 2\pi\, \delta_{2\pi}(t-\gamma_1) + D_1 e' y \sin\varphi\, 2\pi\, \delta_{2\pi}(t-\gamma_2).\end{aligned} \tag{9.25}$$

Wisdom chose $\zeta_1 = \zeta_2 = \zeta_3 = 0$ and $\gamma_1 = \gamma_2 = \pi/2$, which allowed the impulses to act at two different times in the course of one Jupiter period.

Now consider just the secular part of the Hamiltonian. We have

$$\mathcal{H}^{(\mathrm{sec})} = -\frac{\mu_1^2}{2\Phi^2} - 3\Phi + \Lambda' + F_1(x^2+y^2) + \bar{F}_1 e' x, \tag{9.26}$$

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and the equations of motion governing the secular evolution are

$$\begin{aligned} \dot{x} &= -\frac{\partial \mathcal{H}^{(\mathrm{sec})}}{\partial y} = -2F_1 y, & \dot{\Phi} &= -\frac{\partial \mathcal{H}^{(\mathrm{sec})}}{\partial \varphi} = 0, \\ \dot{y} &= +\frac{\partial \mathcal{H}^{(\mathrm{sec})}}{\partial x} = +2F_1 x + \bar{F}_1 e', & \dot{\varphi} &= +\frac{\partial \mathcal{H}^{(\mathrm{sec})}}{\partial \Phi} = \frac{\mu_1^3}{\Phi^3} - 3. \end{aligned} \tag{9.27}$$

The solutions for Φ and φ are trivial:

$$\Phi = \Phi(t_0) = \Phi_0, \qquad \varphi = \left(\frac{\mu_1^2}{\Phi_0^3} - 3\right)(t - t_0) + \varphi_0, \tag{9.28}$$

while the solutions for x and y are

$$\begin{aligned} x &= x(t_0)\,\cos[2F_1(t-t_0)] - y(t_0)\sin[2F_1(t-t_0)] \\ &\quad - \frac{\bar{F}_1 e'}{2F_1}\left[1 - \cos[2F_1(t-t_0)]\right], \\ y &= x(t_0)\,\sin[2F_1(t-t_0)] + y(t_0)\cos[2F_1(t-t_0)] \\ &\quad + \frac{\bar{F}_1 e'}{2F_1}\sin[2F_1(t-t_0)]. \end{aligned} \tag{9.29}$$

Thus, as we saw in Chapter 7, there is an analytical solution for the motion under purely secular perturbations.

Now consider the resonant terms. Consider just the first term where the Hamiltonian is given by

$$\mathcal{H}_1^{(\mathrm{res})} = -\frac{2\pi C_1}{\Delta}(x^2 - y^2)\cos\varphi, \tag{9.30}$$

where

$$\delta(t - t_i) \leftarrow \lim_{\Delta \to 0} \begin{cases} 1/\Delta, & \text{if } t_i \le t < t_i + \Delta, \\ 0, & \text{otherwise.} \end{cases} \tag{9.31}$$

The equations of motion for this particular resonant term are

$$\dot{x} = -\frac{\partial \mathcal{H}_1^{(\mathrm{sec})}}{\partial y} = -\frac{4\pi C_1}{\Delta} y\cos\varphi = -\frac{R_1}{\Delta} y_1, \tag{9.32}$$

$$\dot{y} = +\frac{\partial \mathcal{H}_1^{(\mathrm{sec})}}{\partial x} = -\frac{4\pi C_1}{\Delta} x\cos\varphi = -\frac{R_1}{\Delta} x_1, \tag{9.33}$$

$$\dot{\Phi} = -\frac{\partial \mathcal{H}_1^{(\mathrm{sec})}}{\partial \varphi} = -\frac{2\pi C_1}{\Delta}(x^2 - y^2)\sin\varphi, \tag{9.34}$$

$$\dot{\varphi} = +\frac{\partial \mathcal{H}_1^{(\mathrm{sec})}}{\partial \Phi} = 0, \tag{9.35}$$

where $R_1 = 4\pi C_1 \cos\varphi$ and is a constant since $\dot{\varphi} = 0$ for this Hamiltonian. The solution for x and y is

$$x = x(t_1)\cosh\left[\frac{R_1(t - t_1)}{\Delta}\right] - y(t_1)\sinh\left[\frac{R_1(t - t_1)}{\Delta}\right], \tag{9.36}$$

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$$y = y(t_1)\cosh\left[\frac{R_1(t-t_1)}{\Delta}\right] - x(t_1)\sinh\left[\frac{R_1(t-t_1)}{\Delta}\right]. \tag{9.37}$$

Now set $t = t_1 + \Delta$ and let $\Delta \to 0$. The corresponding changes in the elements are

$$\Delta x = x(t_1)\left(\cosh\left[4\pi C_1 \cos\varphi\right] - 1\right) - y(t_1)\sinh[4\pi C_1 \cos\varphi], \tag{9.38}$$

$$\Delta y = y(t_1)\left(\cosh\left[4\pi C_1 \cos\varphi\right] - 1\right) - x(t_1)\sinh[4\pi C_1 \cos\varphi]. \tag{9.39}$$

Similarly we have

$$\Delta\Phi = -2\pi C_1\left([x(t_1)]^2 - [y(t_1)]^2\right)\sin\varphi(t_1). \tag{9.40}$$

This procedure can be repeated for each of the four other resonant terms given by the Hamiltonians

$$\mathcal{H}_2^{(\mathrm{res})} = -\frac{2\pi D_1}{\Delta}e' x\cos\varphi, \tag{9.41}$$

$$\mathcal{H}_3^{(\mathrm{res})} = -\frac{2\pi E_1}{\Delta}e'^2\cos\varphi, \tag{9.42}$$

$$\mathcal{H}_4^{(\mathrm{res})} = \frac{4\pi C_1}{\Delta}xy\sin\varphi, \tag{9.43}$$

$$\mathcal{H}_5^{(\mathrm{res})} = \frac{2\pi D_1}{\Delta}e' y\sin\varphi. \tag{9.44}$$

For each of these Hamiltonians we can derive the instantaneous changes in the orbital elements just as we have done for the first resonant Hamiltonian, $\mathcal{H}_1^{(\mathrm{res})}$, above.

Let the initial conditions be $x^{(0)}$, $y^{(0)}$, $\Phi^{(0)}$, $\varphi^{(0)}$ at time $t = 0$. The first mapping is applied at $t = 0$ (as determined by the constant ζ_1) and from Eqs. (9.38)–(9.40) we obtain

$$x^{(1)} = x^{(0)}\cosh\left[4\pi C_1\cos\varphi^{(0)}\right] - y^{(0)}\sinh\left[4\pi C_1\cos\varphi^{(0)}\right], \tag{9.45}$$

$$y^{(1)} = y^{(0)}\cosh\left[4\pi C_1\cos\varphi^{(0)}\right] - x^{(0)}\sinh\left[4\pi C_1\cos\varphi^{(0)}\right], \tag{9.46}$$

$$\Phi^{(1)} = \Phi^{(0)} - 2\pi C_1\left(\left[x^{(0)}\right]^2 - \left[y^{(0)}\right]^2\right)\sin\varphi^{(0)}, \tag{9.47}$$

$$\varphi^{(1)} = \varphi^{(0)}. \tag{9.48}$$

Adopting similar procedures for $\mathcal{H}_2^{(\mathrm{res})}$ and $\mathcal{H}_3^{(\mathrm{res})}$, both of which are applied at time $t = 0$, we derive the second and third steps of the mapping:

$$x^{(2)} = x^{(1)}, \tag{9.49}$$

$$y^{(2)} = y^{(1)} - 2\pi D_1 e'\cos\varphi^{(1)}, \tag{9.50}$$

$$\Phi^{(2)} = \Phi^{(1)} - 2\pi D_1 e' x^{(1)}\sin\varphi^{(1)}, \tag{9.51}$$

$$\varphi^{(2)} = \varphi^{(1)} \tag{9.52}$$

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and

$$x^{(3)} = x^{(2)}, \tag{9.53}$$
$$y^{(3)} = y^{(2)}, \tag{9.54}$$
$$\Phi^{(3)} = \Phi^{(2)} - 2\pi E_1 e'^2 \sin\varphi^{(2)}, \tag{9.55}$$
$$\varphi^{(3)} = \varphi^{(2)}. \tag{9.56}$$

The remaining two resonant impulses are applied at time $t = \pi/2$ and so we make use of the analytical solution of the secular part to move the solution ahead to that point. We have $t - t_0 = \pi/2$ and Eqs. (9.28) and (9.29) give us

$$x^{(4)} = x^{(3)} \cos\pi F_1 - y^{(3)} \sin\pi F_1 - \frac{\bar{F}_1 e'}{2F_1}(1 - \cos\pi F_1), \tag{9.57}$$
$$y^{(4)} = x^{(3)} \sin\pi F_1 + y^{(3)} \cos\pi F_1 + \frac{\bar{F}_1 e'}{2F_1} \sin\pi F_1, \tag{9.58}$$
$$\Phi^{(4)} = \Phi^{(3)}, \tag{9.59}$$
$$\varphi^{(4)}\varphi^{(3)} + \frac{\pi}{2}\left(\frac{\mu_1^2}{\left[\Phi^{(3)}\right]^3} - 3\right) \tag{9.60}$$

for the fourth part of the mapping. We can now proceed to apply the impulses due to $\mathcal{H}_4^{(\mathrm{res})}$ and $\mathcal{H}_5^{(\mathrm{res})}$. This gives

$$x^{(5)} = x^{(4)} \exp[+4\pi C_1 \sin\varphi^{(4)}], \tag{9.61}$$
$$y^{(5)} = y^{(4)} \exp[-4\pi C_1 \sin\varphi^{(4)}], \tag{9.62}$$
$$\Phi^{(5)} = \Phi^{(4)} + 4\pi C_1 x^{(4)} y^{(4)} \cos\varphi^{(4)}, \tag{9.63}$$
$$\varphi^{(5)} = \varphi^{(4)} \tag{9.64}$$

and

$$x^{(6)} = x^{(5)} + 2\pi D_1 e' \sin\varphi^{(5)}, \tag{9.65}$$
$$y^{(6)} = y^{(5)}, \tag{9.66}$$
$$\Phi^{(6)} = \Phi^{(5)} + 2\pi D_1 e' y^{(5)} \cos\varphi^{(5)}, \tag{9.67}$$
$$\varphi^{(6)} = \varphi^{(5)}. \tag{9.68}$$

Finally, we apply the solution to the secular part to move the orbit from $t = \pi/2$ to $t = 2\pi$, to ready it for the next application of the $\mathcal{H}_1^{(\mathrm{res})}$ and to allow the next cycle of the mapping to begin. Hence the final mapping is

$$x^{(7)} = x^{(6)} \cos 3\pi F_1 - y^{(6)} \sin 3\pi F_1 - \frac{\bar{F}_1 e'}{2F_1}(1 - \cos 3\pi F_1), \tag{9.69}$$
$$y^{(7)} = x^{(6)} \sin 3\pi F_1 + y^{(6)} \cos 3\pi F_1 + \frac{\bar{F}_1 e'}{2F_1} \sin 3\pi F_1, \tag{9.70}$$
$$\Phi^{(7)} = \Phi^{(6)}, \tag{9.71}$$

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$$\varphi^{(7)} = \varphi^{(6)} + \frac{3\pi}{2}\left(\frac{\mu_1^2}{\left[\Phi^{(6)}\right]^3} - 3\right). \tag{9.72}$$

Wisdom's map relies on (i) the fact that the secular problem is integrable for low eccentricity and inclination (recall that secular terms greater than second order are ignored) and (ii) the replacement of the resonant terms in the disturbing function by impulses applied at specific times. This accounts for the speed of the map, which Wisdom (1982) estimated at 10^3 times faster than solving the full equations of motion. Murray & Fox (1984) compared the behaviour of the map with that derived from an integration of the averaged equations from which the map is derived and a full integration of the equations of motion. They showed that the map gave excellent agreement with the averaged system and only differed from the full system in a quantitative manner. However, the map could be used to predict the regular or chaotic nature of the true orbit, and the map's speed made it a powerful tool for the study of chaotic orbits in the asteroid belt.

Since the 3:1 resonance is a second-order resonance we require terms up to at least order 2 in the disturbing function. Note that the secular terms that contribute to precession only appear at second order and higher. If we wish to study first-order resonances using a map then we should include resonant terms up to at least second order to be consistent with the secular contribution. Murray (1986) took this approach when he used Wisdom's method to derive maps for the 2:1 and 3:2 resonances. As well as the resonant arguments, which contained $2\lambda' - \lambda$ (and $3\lambda' - 2\lambda$), for consistency he also included the second-order arguments which contained $4\lambda' - 2\lambda$ (and $6\lambda' - 4\lambda$). Šidlichovsky & Melendo (1986) derived a similar map for the 5:2 resonance in the asteroid belt. Although they were dealing with a third-order resonance and used second-order secular terms in their derivation, this was still consistent since the higher-order secular terms only come in at order 4 and above. Note that a common problem with all these approaches is the convergence of the disturbing function expansion for large values of $\alpha = a/a'$ (see the discussion in Sect. 9.8.3).

9.5.3 Encounter Maps

Resonance maps use a disturbing function approach to calculate the averaged effect of the perturbation over time. In direct contrast an encounter map approximates the direct effect on the test particle as it encounters a perturber. The technique was employed by Duncan et al. (1989) in their derivation of a map to study the long-term evolution of orbits in the outer solar system. Since we are dealing with perturbed motion of the test particle in the vicinity of the perturber it is natural to choose a coordinate system centred on the perturber. Therefore we can make use of Hill's equations (see Sect. 3.13).

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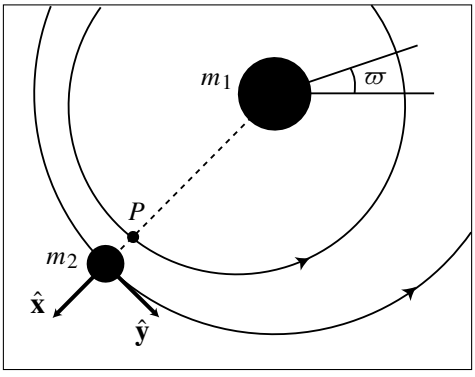


Fig. 9.19. Geometry of the conjunction between a test particle P on a near-circular interior orbit and a perturber on a circular orbit.

Consider the motion of a test particle orbiting a central body of mass m_1. The particle is subject to perturbations from a planet in a circular, coplanar orbit with semi-major axis a' and mass $m_2 \ll m_1$. In a frame centred on the mass m_2 with the x axis pointing radially outwards and the y axis at 90° to it in the direction of motion (see Fig. 9.19) the approximate equations of motion are

$$\ddot{x} - 2n_2\dot{y} - 3n_2^2 x = -\frac{\mathcal{G}m_2 x}{(x^2+y^2)^{3/2}}, \tag{9.73}$$

$$\ddot{y} + 2n_2\dot{x} = -\frac{\mathcal{G}m_2 y}{(x^2+y^2)^{3/2}}, \tag{9.74}$$

where $n_2^2 = \mathcal{G}m_1/a_2^3$ and n_2 denotes the mean motion of the mass m_2; these equations should be compared with Eqs. (3.202) and (3.203). There is also an integral of the motion that is equivalent to the Jacobi integral in the restricted problem. It is given by (cf. Eq. (3.205))

$$C_{\mathrm{H}} = 3n_2^2 x^2 - \dot{x}^2 - \dot{y}^2 + \frac{2\mathcal{G}m_2}{(x^2+y^2)^{3/2}}. \tag{9.75}$$

Hénon & Petit (1986) showed that in the limit $m_2 \to 0$ Hill's equations have the solution

$$x = D_1 \cos n_2 t + D_2 \sin N_2 t + D_3, \tag{9.76}$$

$$y = -2D_1 \sin n_2 t + 2D_2 \cos n_2 t - \frac{3}{2} D_3 n_2 t + D_4, \tag{9.77}$$

where D_1, D_2 D_3, and D_4 are constants of integration. Substituting these values in the equation for C_{H} with $m_2 \to 0$ gives the limiting value of the constant as

$$C_{\mathrm{H}} = n_2^2 \left(\frac{3}{4} D_3^2 - D_1^2 - D_2^2 \right). \tag{9.78}$$

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The constant D_4 can be taken to be zero by choosing the time origin at each conjunction, and D_3 is given by

$$D_3 = a - a_2, \tag{9.79}$$

where a is the semi-major axis of the particle. Hénon and Petit (1986) showed that to lowest order in $|a - a_2|/a_2$ the values of D_1 and D_2 are related to the eccentricity e and longitude of pericentre ϖ by

$$D_1 + \mathrm{i}D_2 = -a_2 z \exp(-\mathrm{i}\lambda_c), \tag{9.80}$$

where λ_c is the longitude of mass m_2 at conjunction and

$$z = e \exp(\mathrm{i}\varpi) \tag{9.81}$$

is a complex eccentricity.

Now we need to calculate the changes in D_1, D_2, and D_3 that occur due to the conjunction. For this step Duncan et al. (1989) followed the work of Julian & Toomre (1966) and Hénon & Petit (1986). For small initial eccentricity the relevant changes are

$$\Delta D_1 = 0, \tag{9.82}$$

$$\Delta D_2 = -g \,\mathrm{sign}\,(D_3) \frac{m_2}{m_1} \frac{a_2^3}{D_3^2}, \tag{9.83}$$

$$\Delta D_3 = 0, \tag{9.84}$$

where the constant g is given by

$$g = \frac{8}{9}\left[2K_0(2/3) + K_1(2/3)\right] = 2.239566674, \tag{9.85}$$

where K_0 and K_1 are modified Bessel functions of the second kind. Therefore the effect of the conjunction is to change the complex eccentricity by an amount

$$\Delta z = \frac{\mathrm{i}g \exp(\mathrm{i}\lambda_c)}{\varepsilon_2} \mathrm{sign}\,(\varepsilon) \frac{m_2}{m_1}, \tag{9.86}$$

where ε is the relative difference in semi-major axis given by

$$\varepsilon = \frac{a - a_2}{a_2}. \tag{9.87}$$

Duncan et al. pointed out that the same result can be derived without using Hill's equations, by means of Gauss's equations or Lagrange's equations.

The fact that ΔD_3 is zero to lowest order implies that there is no change in semi-major axis. However, the modified form of the Jacobi constant given in Eq. (9.78) allows the derivation of a more accurate expression. Let a' and $z' = z + \Delta z$ denote the values of the semi-major axis and complex eccentricity

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immediately after the conjunction with $\varepsilon' = (a' - a_2)/a_2$. Hence, the expression for $C_{\rm H}$ gives

$$\varepsilon'^2 = \varepsilon^2 + \frac{4}{3}\left(|z'|^2 - |z|^2\right). \tag{9.88}$$

Using Eqs. (9.86) and (9.88) we can calculate the values of a' and z' from a and z. We will assume that the new values remain constant until the next conjunction occurs. The time between successive conjunctions (the synodic period) is given by

$$T_{\rm s} = \frac{2\pi}{|n_2 - n|} = \frac{2\pi}{n_2}\left|\left(\frac{a_2}{a'}\right)^{3/2} - 1\right|^{-1} \tag{9.89}$$

and hence the next longitude at conjunction, $\lambda'_{\rm c}$, is given by

$$\lambda'_{\rm c} = \lambda_{\rm c} + 2\pi\ f(\varepsilon'), \tag{9.90}$$

where

$$f(\varepsilon) = \left|(1+\varepsilon)^{-3/2} - 1\right|^{-1}. \tag{9.91}$$

Let λ_n denote the longitude of the nth conjunction and let z_n, and ε_n denote the values of the complex eccentricity and relative semi-major axis just before the nth conjunction. Putting all three changes together we have the complete map as given by Duncan et al. (1989):

$$z_{n+1} = z_n + \frac{{\rm i}g\exp({\rm i}\lambda_n)}{\varepsilon_1}\,{\rm sign}(\varepsilon_1)\frac{m_2}{m_1}, \tag{9.92}$$

$$\varepsilon_{n+1} = \varepsilon_n\sqrt{1 + \frac{4\left(|z_{n+1}|^2 - |z_n|^2\right)}{3\varepsilon_n^2}}, \tag{9.93}$$

$$\lambda_{n+1} = \lambda_n + 2\pi\ f(\varepsilon_{n+1}). \tag{9.94}$$

Note the occurrence of ε_1 rather than ε_n in Eq. (9.92). The change, which has little effect on the accuracy of the map, was made by Duncan et al. in order to make the map area preserving.

Duncan et al. (1989) give a number of alternative forms of the map, each of which illustrates a particular property of the system. By introducing $\Delta\varepsilon_n = \varepsilon_n - \varepsilon_1$, considering the case where $\varepsilon_1 > 0$, and expanding the right hand sides of Eqs. (9.93) and (9.94), we obtain

$$z_{n+1} = z_n + \frac{{\rm i}g\exp({\rm i}\lambda_n)}{\varepsilon_1}\frac{m_2}{m_1}, \tag{9.95}$$

$$\Delta\varepsilon_{n+1} = \frac{2\left(|z_{n+1}|^2 - |z_n|^2\right)}{3\varepsilon_1}, \tag{9.96}$$

$$\lambda_{n+1} = \lambda_n + \frac{4\pi}{3\varepsilon_1} - \frac{4\pi\,\Delta\varepsilon_{n+1}}{3\varepsilon_1^2}. \tag{9.97}$$

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The map derived by Duncan et al. was also modified to include the effect of a perturber moving in an elliptical orbit. They showed that all that was required was to replace the original definition of z by

$$z = e\exp(\mathrm{i}\varpi) - e_2\exp(\mathrm{i}\varpi_2), \tag{9.98}$$

where e_2 and ϖ_2 are the eccentricity and longitude of pericentre respectively of the perturbing body. Everything else remains the same. In a further modification Duncan et al. showed how the map could be adapted to include the effect of perturbations from two planets on circular orbits.

Namouni et al. (1996) included higher order terms in the eccentricity and inclination in the expansion of the interaction, leading to an improved mapping for encounters in the Hill's approximation. In all of these cases, as with the resonance maps, the power of the encounter map lies in its ability to "jump" instantly to the next perturbation by means of an analytical solution. We will discuss the uses of the map in Sect. 9.6.

9.5.4 *N*-Body Maps

The addition of an infinite number of high-frequency terms to the Hamiltonian was a necessary part of the derivation of the resonance maps. These maps also rely on truncated expansions of the planetary disturbing function and its inherent problems. Wisdom & Holman (1991) derived a map for the N-body problem that made use of the trick involving the Dirac δ-function but avoided the need for series expansions in orbital elements. Wisdom & Holman's map is an important tool because of (i) its speed and (ii) its *symplectic* nature, which results in no secular changes in the energy of the system. The use of this map was to have a profound effect on studies of long-term stability in the solar system and its derivation is worth examining in some detail.

The starting point for the N-body map is the Hamiltonian of the gravitational N-body problem where we are required to find the solution for a system of N mutually perturbed bodies. Let the ith body have a mass m_i, a linear momentum $\mathbf{p}_i = m_i\mathbf{v}_i$, and a separation from the jth body of $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$. The Hamiltonian of the system is given by (cf. Eq. (2.170))

$$\mathcal{H} = \sum_{j=0}^{N-1}\frac{p_j^2}{2m_j} - \sum_{j=0}^{N-2}\sum_{k=j+1}^{N-1}\frac{\mathcal{G}m_jm_k}{r_{jk}}. \tag{9.99}$$

To derive an N-body map this Hamiltonian has to be separated into a keplerian part and an interaction part (Wisdom & Holman 1991). A Hamiltonian is defined to be keplerian (see Sect. 2.10) if it is possible to write it in the form

$$\mathcal{H} = \frac{p^2}{2m} - \frac{\mathcal{G}Mm}{r} \tag{9.100}$$

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or as the sum of such forms. In order to achieve the proper separation of the Hamiltonian, it is necessary to choose a new system of variables called the *Jacobian coordinates* (see, for example, Plummer 1918).

Consider $N-1$ bodies of mass $m_1, m_2, \ldots, m_{N-1}$ orbiting a central body of mass m_0. With respect to a fixed origin at O the N bodies have position vectors $\mathbf{r}_0, \mathbf{r}_1, \ldots, \mathbf{r}_{N-1}$ (see Fig. 9.20a for the case where $N = 4$). Let $\mathbf{R}_0$ denote the position vector of the centre of mass of a system consisting of the central mass alone (i.e., $\mathbf{R}_0 = \mathbf{r}_0$) with $\mathbf{R}_1$ denoting the centre of mass of the system composed of the masses m_0 and m_1, and so on. Hence $\mathbf{R}_i$ denotes the position vector of the centre of mass of all the bodies with index up to and including i (see Fig. 9.20b). We can write this definition as

$$\mathbf{R}_i = \frac{1}{\eta_i} \sum_{j=0}^{i} m_j \mathbf{r}_j, \tag{9.101}$$

where

$$\eta_i = \sum_{j=0}^{i} m_i. \tag{9.102}$$

The first Jacobian position vector, $\tilde{\mathbf{r}}_0$ (the tilde denotes a value in the Jacobian system), is the position vector of the centre of mass of the whole system. Hence

$$\tilde{\mathbf{r}}_0 = \mathbf{R}_{N-1} \tag{9.103}$$

for our N-body system of $N-1$ bodies orbiting the central mass. The remaining Jacobian position vectors are given by

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i - \mathbf{R}_{i-1}. \tag{9.104}$$

Hence $\tilde{\mathbf{r}}_1$ denotes the position vector of m_1 with respect to m_0, $\tilde{\mathbf{r}}_2$ denotes the position vector of m_2 with respect to the centre of mass of m_0 and m_1, etc. (see Fig. 9.20b).

In the Jacobian coordinate system the momentum vectors conjugate to the position vectors $\tilde{\mathbf{r}}_i$ are

$$\tilde{\mathbf{p}}_i = \tilde{m}_i \, \tilde{\mathbf{v}}_i, \tag{9.105}$$

where $\tilde{\mathbf{v}}_i = \dot{\tilde{\mathbf{r}}}_i$ and the $\tilde{m}_i$ are given by

$$\tilde{m}_i = \begin{cases} (\eta_{i-1}/\eta_i)\, m_i, & \text{if } 0 < i < N, \\ \eta_{N-1} = M_{\text{total}}, & \text{if } i = 0, \end{cases} \tag{9.106}$$

where M_{total} is the total mass of the system. Expressions for the Jacobian momentum vectors in terms of their Cartesian counterparts can be found by differentiating Eqs. (9.103) and (9.104) and multiplying by $\tilde{m}_0$ and $\tilde{m}_i$ respectively.

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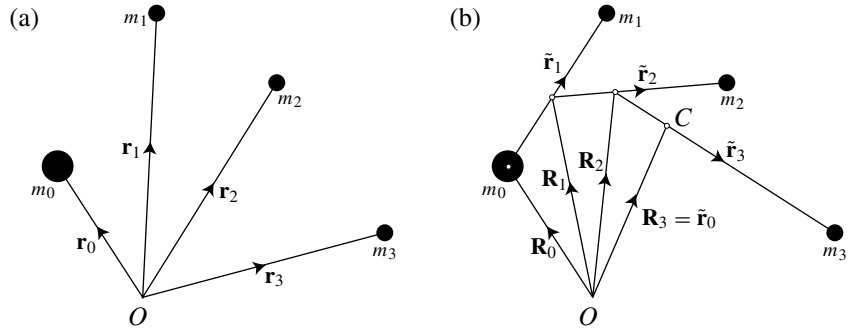


Fig. 9.20. The Jacobian coordinate system for a four-body problem where the position of the ith body is referred to the centre of mass of the system of the previous $i-1$ bodies. (a) The position vectors $\mathbf{r}_i$ of four masses with respect to the origin at O. (b) The position vectors $\mathbf{R}_i$ of the centres of mass of the different subsystems (denoted by small open circles) and the position vectors $\tilde{\mathbf{r}}_i$ in the Jacobian coordinate system relative to the centres of mass for the same system. C denotes the centre of mass of the system.

This gives

$$\tilde{\mathbf{p}}_0 = \sum_{j=0}^{N-1} \mathbf{p}_j, \tag{9.107}$$

which is the total linear momentum of the system, and

$$\tilde{\mathbf{p}}_i = \left(\frac{\eta_{i-1}}{\eta_i}\right)\mathbf{p}_i - \left(\frac{m_i}{\eta_i}\right)\sum_{j=0}^{i-1}\mathbf{p}_j \qquad (0 < i < N) \tag{9.108}$$

(see Saha & Tremaine 1994).

To express $\mathcal{H}$ in terms of the Jacobian coordinates and momenta we need to find suitable expressions that relate the standard Cartesian system to the Jacobian system. By writing out expressions for each of the $\tilde{\mathbf{r}}_i$, multiplying each by η_{i-1}, and subtracting consecutive pairs of equations, it can be shown (Tom Kehoe, private communication) that the standard coordinates can be written as

$$\mathbf{r}_0 = \tilde{\mathbf{r}}_0 - \sum_{j=1}^{N-1}\left(\frac{m_j}{\eta_j}\right)\tilde{\mathbf{r}}_j, \tag{9.109}$$

$$\mathbf{r}_i = \tilde{\mathbf{r}}_0 + \left(\frac{\eta_{i-1}}{\eta_i}\right)\tilde{\mathbf{r}}_i - \sum_{j=i+1}^{N-1}\left(\frac{m_j}{\eta_j}\right)\tilde{\mathbf{r}}_j \qquad (0 < i < N-1), \tag{9.110}$$

$$\mathbf{r}_{N-1} = \tilde{\mathbf{r}}_0 + \left(\frac{\eta_{N-2}}{\eta_{N-1}}\right)\tilde{\mathbf{r}}_{N-1}. \tag{9.111}$$

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These equations can be used to derive the following expressions for the relative position vectors:

$$\mathbf{r}_{i0} = \tilde{\mathbf{r}}_i + \sum_{j=1}^{i-1} \left(\frac{m_j}{\eta_j} \right) \tilde{\mathbf{r}}_j \qquad (0 < i < N), \tag{9.112}$$

$$\mathbf{r}_{ij} = \left(\frac{\eta_{i-1}}{\eta_i} \right) \tilde{\mathbf{r}}_i - \sum_{k=i+1}^{j-1} \left(\frac{m_k}{\eta_k} \right) \tilde{\mathbf{r}}_k - \tilde{\mathbf{r}}_j \qquad (0 < i < j < N). \tag{9.113}$$

By differentiating Eqs. (9.109)–(9.111) with respect to time and multiplying them by m_0, m_i, and m_{N-1} respectively, we derive the following expressions for the corresponding momenta:

$$\mathbf{p}_0 = \left(\frac{m_0}{\eta_{N-1}} \right) \tilde{\mathbf{p}}_0 - \sum_{j=1}^{N-1} \left(\frac{m_0}{\eta_{j-1}} \right) \tilde{\mathbf{p}}_j, \tag{9.114}$$

$$\mathbf{p}_i = \left(\frac{m_i}{\eta_{N-1}} \right) \tilde{\mathbf{p}}_0 + \tilde{\mathbf{p}}_i - \sum_{j=i+1}^{N-1} \left(\frac{m_i}{\eta_{j-1}} \right) \tilde{\mathbf{p}}_j \qquad (0 < i < N-1), \tag{9.115}$$

$$\mathbf{p}_{N-1} = \left(\frac{m_{N-1}}{\eta_{N-1}} \right) \tilde{\mathbf{p}}_0 + \tilde{\mathbf{p}}_{N-1}. \tag{9.116}$$

With these definitions the kinetic energy term of the Hamiltonian in Eq. (9.99) can now be written as

$$\sum_{j=0}^{N-1} \frac{p_j^2}{2m_j} = \frac{\tilde{p}_0^2}{2\tilde{m}_0} + \sum_{j=1}^{N-1} \frac{\tilde{p}_j^2}{2\tilde{m}_j}. \tag{9.117}$$

Plummer (1918) gives an alternative derivation of this expression without the need for the intermediate expressions for the momentum vectors. In terms of the Jacobian coordinates the full Hamiltonian is

$$\mathcal{H} = \frac{\tilde{p}_0^2}{2\tilde{m}_0} + \sum_{j=1}^{N-1} \frac{\tilde{p}_j^2}{2\tilde{m}_j} - \sum_{j=0}^{N-2} \sum_{k=j+1}^{N-1} \frac{\mathcal{G} m_j m_k}{r_{jk}}. \tag{9.118}$$

Because r_{ij} does not depend on $\tilde{\mathbf{r}}_0$ (the position vector of the centre of mass), the total Jacobian momentum $\tilde{\mathbf{p}}_0$ has to be an integral of the motion. This is simply another way of stating the well-known result that in the gravitational N-body system with no external forces the centre of mass is either at rest or is moving with uniform velocity in a straight line. Consequently, we can ignore the first term in Eq. (9.118). Adding and subtracting the quantity $\sum_{j=1}^{N-1} \mathcal{G} m_0 m_j / \tilde{r}_j$ to our expression for $\mathcal{H}$ means that we can now write the Hamiltonian as

$$\mathcal{H} = \mathcal{H}_{\text{Kepler}} + \mathcal{H}_{\text{interaction}}, \tag{9.119}$$

where

$$\mathcal{H}_{\text{Kepler}} = \sum_{j=1}^{N-1} \left(\frac{\tilde{p}_j^2}{2\tilde{m}_j} - \frac{\mathcal{G} \tilde{M}_j \tilde{m}_j}{\tilde{r}_j} \right), \tag{9.120}$$

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$$\mathcal{H}_{\text{interaction}} = \sum_{j=1}^{N-1} \frac{\mathcal{G}m_0 m_j}{\tilde{r}_j} - \sum_{j=0}^{N-2} \sum_{k=j+1}^{N-1} \frac{\mathcal{G}m_j m_k}{r_{jk}}, \tag{9.121}$$

where, in $\mathcal{H}_{\text{Kepler}}$, we have expressed the quantity $\sum_{i=j}^{N-1} \mathcal{G}m_0 m_j/\tilde{r}_j$ in terms of the Jacobian masses $\tilde{m}_i$ and $\tilde{M}_i = (\eta_i/\eta_{i-1})\, m_0$ for $0 < i < N$. Now we can consider the Kepler and interaction parts of $\mathcal{H}$ separately.

Note that $\mathcal{H}_{\text{Kepler}}$ depends on the individual $\tilde{r}_i$ and $\tilde{p}_i$. Thus $\mathcal{H}_{\text{Kepler}}$ has now been expressed as the sum of $N-1$ individual two-body problem Hamiltonians as desired. Generalising the expressions given in Eq. (2.170) we have

$$\left(\dot{\tilde{\mathbf{r}}}_i\right)_{\text{Kepler}} = +\nabla_{\tilde{\mathbf{p}}_i} \mathcal{H}_{\text{Kepler}}, \qquad \left(\dot{\tilde{\mathbf{p}}}_i\right)_{\text{Kepler}} = -\nabla_{\tilde{\mathbf{r}}_i} \mathcal{H}_{\text{Kepler}} \tag{9.122}$$

for the evolution of the Jacobian position and momentum vectors under $\mathcal{H}_{\text{Kepler}}$ alone. Hence

$$\left(\dot{\tilde{\mathbf{r}}}_i\right)_{\text{Kepler}} = \frac{\mathbf{p}_i}{\tilde{m}_i}, \qquad \left(\dot{\tilde{\mathbf{p}}}_i\right)_{\text{Kepler}} = -\frac{\mathcal{G}\tilde{M}_i \tilde{m}_i}{\tilde{r}_i^3} \tilde{\mathbf{r}}_i \tag{9.123}$$

for all $0 < i < N$ (cf. Eq. (2.174)). Equations (9.123) combine to give the equation of motion of the ith Jacobian body (of mass $\tilde{m}_i$) under the effect of $\mathcal{H}_{\text{Kepler}}$ as

$$\frac{\mathrm{d}^2 \tilde{\mathbf{r}}_i}{\mathrm{d}t^2} + \frac{\mathcal{G}\tilde{M}_i}{\tilde{r}_i^3} \tilde{\mathbf{r}}_i = \mathbf{0} \tag{9.124}$$

for all $0 < i < N$ (cf. Eq. (2.168)). This can be interpreted as the equation of motion of a body about a fixed mass $\tilde{M}_i$ located at the centre of mass, $\mathbf{R}_{i-1}$, of the previous $i-1$ bodies. It is also the equation of relative motion of two bodies with total mass $\tilde{M}_i$ and vector separation $\tilde{\mathbf{r}}_i$.

The first point to note about $\mathcal{H}_{\text{interaction}}$ is that it depends on neither $\mathbf{p}_i$ nor $\tilde{\mathbf{p}}_i$ and hence, from Hamilton's equations, $\mathbf{r}_i$ and $\tilde{\mathbf{r}}_i$ are constants of the motion under the effect of $\mathcal{H}_{\text{interaction}}$. The next point to note is that $\mathcal{H}_{\text{interaction}}$ can itself be expressed as a sum. We can write

$$\mathcal{H}_{\text{interaction}} = \mathcal{H}_{\text{Jacobian}} + \mathcal{H}_{\text{Cartesian}}, \tag{9.125}$$

where

$$\mathcal{H}_{\text{Jacobian}} = \sum_{j=1}^{N-1} \frac{\mathcal{G}m_0 m_j}{\tilde{r}_j}, \qquad \mathcal{H}_{\text{Cartesian}} = -\sum_{j=0}^{N-2} \sum_{k=j+1}^{N-1} \frac{\mathcal{G}m_j m_k}{r_{jk}}. \tag{9.126}$$

Therefore

$$\dot{\tilde{\mathbf{p}}}_i = -\nabla_{\tilde{\mathbf{r}}_i} \mathcal{H}_{\text{Jacobian}} - \nabla_{\tilde{\mathbf{r}}_i} \mathcal{H}_{\text{Cartesian}} = \left(\dot{\tilde{\mathbf{p}}}_i\right)_{\text{Jacobian}} + \left(\dot{\tilde{\mathbf{p}}}_i\right)_{\text{Cartesian}}, \tag{9.127}$$

where it is easy to show that

$$\left(\dot{\tilde{\mathbf{p}}}_i\right)_{\text{Jacobian}} = \frac{\mathcal{G}m_0 m_i}{\tilde{r}_i^3} \tilde{\mathbf{r}}_i. \tag{9.128}$$

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To calculate $\dot{\tilde{\mathbf{p}}}_i$ under the effect of the Cartesian part of the Hamiltonian, we first find $\dot{\mathbf{p}}_i$ under the same Hamiltonian and then substitute the resulting expression in the time derivative of Eq. (9.108). This procedure gives

$$(\dot{\mathbf{p}}_i)_{\text{Cartesian}} = -\nabla_{\mathbf{r}_i}\mathcal{H}_{\text{Cartesian}} = \sum_{\substack{k=0\\(k\neq i)}}^{N-1} \frac{\mathcal{G}m_i m_k}{r_{ik}^3}\mathbf{r}_{ik}, \tag{9.129}$$

$$\dot{\tilde{\mathbf{p}}}_i = \left(\frac{\eta_{i-1}}{\eta_i}\right)\dot{\mathbf{p}}_i - \left(\frac{m_i}{\eta_i}\right)\sum_{j=0}^{i-1}\dot{\mathbf{p}}_j \tag{9.130}$$

and hence

$$\left(\dot{\tilde{\mathbf{p}}}_i\right)_{\text{Cartesian}} = \left(\frac{\eta_{i-1}}{\eta_i}\right)\sum_{\substack{k=0\\(k\neq i)}}^{N-1}\frac{\mathcal{G}m_i m_k}{r_{ik}^3}\mathbf{r}_{ik} - \left(\frac{m_i}{\eta_i}\right)\sum_{j=0}^{i-1}\sum_{\substack{k=0\\(k\neq j)}}^{N-1}\frac{\mathcal{G}m_j m_k}{r_{jk}^3}\mathbf{r}_{jk} \tag{9.131}$$

for all $0 < i < N$. Combining Eqs. (9.128) and (9.131) and dividing through by $\tilde{m}_i$ and using $\tilde{M}_i = (\eta_i/\eta_{i-1})m_0$ gives

$$\left(\dot{\tilde{\mathbf{v}}}_i\right)_{\text{interaction}} = \frac{\mathcal{G}\tilde{M}_i}{\tilde{r}_i^3}\tilde{\mathbf{r}}_i + \sum_{\substack{k=0\\(k\neq i)}}\frac{\mathcal{G}m_k}{r_{ik}^3}\mathbf{r}_{ik} - \frac{1}{\eta_{i-1}}\sum_{j=0}^{i-1}\sum_{\substack{k=0\\(k\neq j)}}^{N-1}\frac{\mathcal{G}m_j m_k}{r_{jk}^3}\mathbf{r}_{jk}. \tag{9.132}$$

It can be shown (Tom Kehoe, private communication) that many of the terms on the right-hand side of Eq. (9.132) cancel out and the equation simplifies to

$$\begin{aligned}\left(\dot{\tilde{\mathbf{v}}}_i\right)_{\text{interaction}} = \mathcal{G}\tilde{M}_i\left[\frac{\tilde{\mathbf{r}}_i}{\tilde{r}_i^3} - \frac{\mathbf{r}_{0i}}{r_{0i}^3}\right] - \left(\frac{\eta_i}{\eta_{i-1}}\right)\sum_{j=1}^{i-1}\frac{\mathcal{G}m_j}{r_{ji}^3}\mathbf{r}_{ji}\\ + \sum_{j=i+1}^{N-1}\frac{\mathcal{G}m_j}{r_{ij}^3}\mathbf{r}_{ij} - \frac{1}{\eta_{i-1}}\sum_{j=0}^{i-1}\sum_{k=i+1}^{N-1}\frac{\mathcal{G}m_j m_k}{r_{jk}^3}\mathbf{r}_{jk}.\end{aligned} \tag{9.133}$$

Care must be taken when evaluating the term in square brackets because of the numerical dangers of subtracting nearly equal quantities. The final part of taking account of the interaction Hamiltonian is the most trivial. In a time step Δt the Jacobian velocity vector of the ith body changes by an amount $\Delta\tilde{\mathbf{v}}_i$ given by

$$\Delta\tilde{\mathbf{v}}_i = \Delta t\left(\dot{\tilde{\mathbf{v}}}_i\right)_{\text{interaction}} \qquad (0 < i < N) \tag{9.134}$$

while all the $\tilde{\mathbf{r}}_i$ and $\mathbf{r}_i$ remain constant.

The speed of a map comes from its ability to make use of an integrability (or an approximation to it) within the system. In the case of resonance maps it was the fact that the ever-present secular perturbations are integrable and the resonant perturbations are applied as impulses. For the encounter map it is assumed that there is unperturbed motion between the successive impulses at conjunction. In the case of the Wisdom & Holman N-body map we exploit the essentially Keplerian nature of the orbit. The Hamiltonian is now in a form suitable for the addition of a single δ-function and the derivation of a map. This can be achieved

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in a similar way to that discussed for the resonance maps in Sect. 9.5.2. The basis of Wisdom & Holman's N-body map is to replace the Hamiltonian given in Eq. (9.119) by

$$\mathcal{H} = \mathcal{H}_{\text{Kepler}} + 2\pi\,\delta_{2\pi}(\Omega t)\mathcal{H}_{\text{interaction}}, \tag{9.135}$$

where $\delta_{2\pi}(\Omega t)$ is a 2π-periodic Dirac δ-function with mapping frequency Ω; this use of δ-functions is identical to that discussed in Sect. 9.5.2 for resonance maps, although in that case $\Omega = 1$. Note again that because the $\mathcal{H}_{\text{interaction}}$ is independent of the momenta, the coordinates remain constant over the application of the impulse and it is only the momenta that change. Between applications of the δ-functions (associated with evolution under $\mathcal{H}_{\text{interaction}}$ given in Eq. (9.134)) the system is stepped forward to the next jump by means of the f and g series (associated with the evolution under $\mathcal{H}_{\text{Kepler}}$ given by Eq. (9.124)). This is based on the principle that at any time t_0 the position vector $\mathbf{r}$ and the velocity vector $\mathbf{v}$ determine an orbital plane and so the values of $\mathbf{r}$ and $\mathbf{v}$ at time t can be written as the linear combination

$$\mathbf{r}(t) = f(t)\,\mathbf{r}(t_0) + g(t)\,\mathbf{v}(t_0) \tag{9.136}$$

and hence

$$\mathbf{v}(t) = \dot{f}(t)\,\mathbf{r}(t_0) + \dot{g}(t)\,\mathbf{v}(t_0). \tag{9.137}$$

The functions f, g, $\dot{f}$, and $\dot{g}$ are defined in Eqs. (2.69) and (2.71), each of which involves the difference in eccentric anomalies, $\Delta E = E - E_0$. The difference form of Kepler's equation can be written

$$n\,\Delta t = \Delta E - e\cos E_0 \sin\Delta E + e\sin E_0(1 - \cos\Delta E) \tag{9.138}$$

and this has to be solved for each mass to advance the system to the next impulse. Using f and g functions the Kepler step and the interaction step can both be evolved using Cartesian coordinates, which, as Wisdom & Holman point out, avoids the need for computationally expensive coordinate transformations between steps (Wisdom & Holman 1991). Methods for the solution of Kepler's equation are discussed in Sect. 2.4.

Wisdom & Holman point out that there are two choices regarding optimization of the map. Either one can choose to have the Hamiltonian of the map and the true Hamiltonian to agree as much as possible or, alternatively, one can have agreement in the Taylor series solution for the real system and the map over one time step. Let the Hamiltonian for the map have the form

$$\mathcal{H}_{\text{Map}} = \mathcal{H}_{\text{Kepler}} + \Phi(\Omega t)\mathcal{H}_{\text{interaction}}, \tag{9.139}$$

where, because the central mass is usually much larger than the perturbing mass, $\mathcal{H}_{\text{Kepler}} \gg \mathcal{H}_{\text{interaction}}$, and

$$\Phi(t) = 2\pi \sum_{i=0}^{k-1} a_i\,\delta_{2\pi}(t - 2\pi d_i). \tag{9.140}$$

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This means that there are k δ-functions for each mapping period. The functions have amplitudes a_i and phases d_i, where $0 < d_i < 1$. In terms of Fourier series,

$$\Phi(t) = \sum_{i=0}^{k-1} a_i \sum_{n=-\infty}^{\infty} \cos[n(t - 2\pi d_i)] \tag{9.141}$$

$$= \sum_{n=-\infty}^{\infty} A_n \cos(nt) + \sum_{n=-\infty}^{\infty} B_n \sin(nt), \tag{9.142}$$

where

$$A_n = \sum_{i=0}^{k-1} a_i \cos(2\pi n d_i), \qquad B_n = \sum_{i=0}^{k-1} a_i \cos(2\pi n d_i). \tag{9.143}$$

If we want to get the best agreement between the two Hamiltonians then, for a given value of k (the order of the map), we have to find the values of the $2k$ quantities a_i and d_i that optimise the agreement with the true Hamiltonian. We can do this by placing conditions on the A_i and B_i. Since the averaged value of the mapping Hamiltonian over one mapping period has to equal the true Hamiltonian, this implies

$$A_0 = \sum_{i=0}^{k-1} a_i = 1. \tag{9.144}$$

Similarly, for the two Hamiltonians to agree we must have

$$A_n = \sum_{i=0}^{k-1} a_i \cos(2\pi n d_i) = 0, \qquad B_n = \sum_{i=0}^{k-1} a_i \cos(2\pi n d_i) = 0 \tag{9.145}$$

for $n \neq 0$. Wisdom & Holman point out that suitable solutions to these equations for a given order k are $a_i = 1/k$ with the d_i having equal separations. Note that no useful constraint is placed on the absolute phase; all that is required is that the phases are equally spaced.

The alternative approach is to optimise agreement between the series solutions. However, Wisdom & Holman showed that this can be achieved for a second-order (i.e., error is $\mathcal{O}(\Delta t^3)$) map by taking $a_0 = 1$ with $d_0 = 1/2$ and $d_1 = 1$, which simultaneously satisfies the condition for agreement of the Hamiltonians. Let $\Delta t = 2\pi/\Omega$ denote the time interval over which one complete cycle of the map is to be applied. In the "generalised leapfrog" application proposed by Wisdom & Holman all the bodies in the system are evolved as independent two-body Kepler orbits for a time interval $\frac{1}{2}\Delta t$. The impulses from the perturbing planets are then applied, leading to the changes in the Jacobian velocity vectors, $\Delta\tilde{\mathbf{v}}_i$, given by Eq. (9.134). This is followed by evolving all bodies on independent Kepler orbits again for the remaining time $\frac{1}{2}\Delta t$ to complete the time step.

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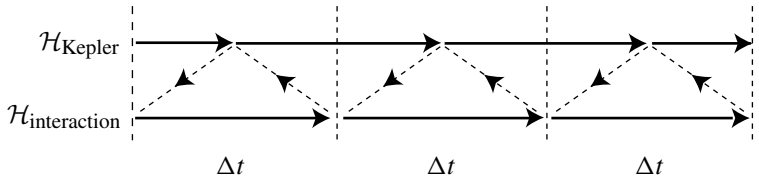


Fig. 9.21. Schematic illustration of the generalised leapfrog method applied to the $k = 1$ case for three equal time intervals, Δt. Solid lines represent evolution under the particular Hamiltonian; dashed lines show the sequence of application.

Figure 9.21 schematically illustrates the continued application of this process for the case $k = 1$. Note that the $\mathcal{H}_{\rm interaction}$ steps (i.e., those involving the application of the δ-functions) always involve equally spaced whole intervals whereas the $\mathcal{H}_{\rm Kepler}$ steps always start and end with a half interval. The map proposed by Wisdom & Holman satisfies the best of both worlds. This is because it preserves the condition $\sum a_i = 1$ while having the δ-functions evenly spaced; furthermore the use of an absolute phase allows it to be an accurate representation of the solution to second order. In fact, it can be shown that the map follows exactly the dynamics of a surrogate Hamiltonian that differs from $\mathcal{H}$ by terms of $\mathcal{O}(\Delta t^2)$. This ensures that there is no secular trend in the energy of the evolved system (Saha & Tremaine 1994).

The use of symplectic integrators of the type developed by Wisdom & Holman has now become widespread. Higher-order methods have been derived (see, for example, Kinoshita et al. 1991) and there have been other enhancements such as the reduction in long-term errors (Saha & Tremaine 1992), the introduction of separate time steps for each body (Saha & Tremaine 1994), a symplectic algorithm that allows for close approaches between bodies (Levison & Duncan 1994), and one that allows for dissipation (Malhotra 1994).

9.6 Separatrices and Resonance Overlap

We have already met the concept of a separatrix in Sect. 8.6 where a particular choice of energy in the pendulum model led to motion with infinite period. Lower values of the energy led to libration and higher values to circulation (see Fig. 8.6b). We have also seen how the pendulum model is a good approximation to motion near resonance in the averaged problem. In both cases the problem is integrable. However, the addition of short-period terms to the pendulum model, leading to the standard map, showed that chaotic motion can arise. A similar type of behaviour is seen in the surfaces of section of the planar restricted three-body problem. In both cases the chaos is associated with motion near the separatrix. Orbits well inside the separatrix and close to the island centres are regular. The chaotic orbits are essentially unpredictable, although they are derived from a deterministic system. Why does motion at the separatrix in the perturbed

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system produce such chaotic orbits? To answer this we will consider a perturbed version of the separatrix diagram originally shown for the unperturbed pendulum in Fig. 8.6b.

Consider Fig. 9.22 where we have plotted some of the curves of constant Hamiltonian for the perturbed case. The unstable equilibrium point can be thought of as being made up of two branches of the curve. The $\mathcal{H}^+$ branches indicate motion towards the unstable point, whereas the $\mathcal{H}^-$ curves indicate motion away from the point. It is the combination of these two branches that leads to near-hyperbolic motion very close to the unstable equilibrium point. However, since the left-hand unstable point is the same as the right-hand unstable point (the angular coordinate is 2π-periodic), the $\mathcal{H}^-$ branch emanating from the left-hand point eventually has to meet the $\mathcal{H}^+$ branch entering the right-hand point. In the unperturbed case the meeting is smooth, but in the perturbed case things are different.

Consider a point x at a crossing point of the two branches. If we consider the motion of the perturbed system in terms of a map F, for example the standard map discussed above, then after one application of the map the new value is $x' = F(x)$, after two applications it is $x'' = F(x') = F^2(x)$, etc. Hence if x is on the $\mathcal{H}^+$ branch the unstable point is defined by $\lim_{n\to\infty} F^n(x)$, where $F^n(x)$ denotes the nth application of the map F. Because we are dealing with motion on the separatrix we can expect the points to get closer together as $n \to \infty$. However, the unstable point can also be thought of as the $\lim_{n\to\infty} T^{-n}x$ if we are considering motion on the $\mathcal{H}^-$ branch. Since we are dealing with a Hamiltonian system the motion is area preserving. Thus, when the curves from the different branches

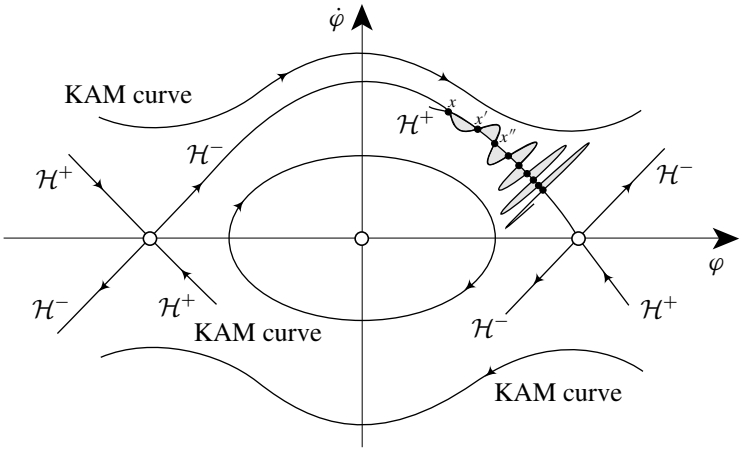


Fig. 9.22. The stable ($\mathcal{H}^+$) and unstable ($\mathcal{H}^-$) branches of the separatrix intersect to generate chaotic motion near the separatrix for a perturbed pendulum. The equilibrium points are denoted by unfilled circles. (Adapted from Lichtenberg & Lieberman 1983.)

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meet, the area enclosed by their intersection (the shaded areas in Fig. 9.22) has to remain constant. But to reconcile this with the fact that the iterated points are getting closer together, we must have the $\mathcal{H}^-$ branch of the curve oscillating more and more wildly, as shown in Fig. 9.22. Eventually, as the unstable point is approached the oscillation will be nearly parallel to the final $\mathcal{H}^-$ curves coming from the unstable point.

Recall also that the period of the unperturbed motion tends to ∞ as the separatrix is approached. This gives rise to numerous resonances with the other frequencies. For example, in Fig. 9.14 we saw how large-amplitude librations at the 5:2 resonance have a frequency that matches the circulation frequency of the 60:24 resonance, producing another eight islands within each of the three main islands. Of course, each of these new islands will have its own chaotic separatrix. It is this mechanism combined with the complicated behaviour of the Hamiltonian curves that gives rise to the chaotic separatrix. The extent of the chaotic region for a given resonance can be determined using methods originally developed by Melnikov (1963) (see, for example, Lichtenberg & Lieberman (1983) and Guckenheimer & Holmes (1983)).

In Fig. 9.18 we noted that apart from the chaotic motion near the separatrix there were still regular orbits left, although they were becoming rare as k_0 increased. The same was also true in the surfaces of section plots for the circular restricted problem (see Figs. 9.16 and 9.17). These are represented by the curves labelled "KAM curves" in Fig. 9.22. (The initials refer to Kolmogorov, Arnold, and Moser who derived an important result on the continued existence of regular orbits as an integrable system became subject to a small perturbation.)

Chaotic motion can also be thought of as arising from the overlap of adjacent resonances. Consider the case of the circular restricted three-body problem. If we consider the averaged problem in the vicinity of a resonance then, according to the theory we have developed in Chapter 8, there is a well-defined separatrix and maximum width to the resonance. As a first approximation we could consider the phase space as being made up of a succession of such resonances, each independent of the others with no short-period terms to affect the separatrix. An obvious example is the sequence of first-order interior resonances of the form $p+1:p$. In Sect. 8.7 we derived expressions for the maximum libration width of such resonances, each treated in isolation. Since each resonance has a well-defined width in semi-major axis, and since the separation of adjacent resonances becomes smaller as the perturber is approached, there will come a point at which these overlap. Wisdom (1980) showed that in the circular restricted problem for small eccentricity ($e \leq 0.15$) of the particle this point is reached when

$$p_{\text{overlap}} \approx 0.51 \mu_2^{-2/7}, \tag{9.146}$$

where $\mu_2 = m_2/(m_1+m_2)$ is the mass of the perturbing secondary in dimensionless units. Using Kepler's third law we find that this corresponds to a separation

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in semi-major axis of

$$\Delta a_{\text{overlap}} \approx 1.3 \mu_2^{2/7} a', \tag{9.147}$$

where a' is the semi-major axis of the perturber. Therefore we would expect particles in the region $a' \pm \Delta a_{\text{overlap}}$ to be on chaotic orbits, having close encounters with the perturber and being removed from the zone. Note that this result predicts a scaling of $\Delta a_{\text{overlap}}$ as the two-sevenths power of the mass ratio of the system.

Duncan et al. (1989) give a simple heuristic derivation of the same result, using the properties of the encounter map discussed in Sect. 9.5.3. They note that in their system chaos occurs when the perturbation in semi-major axis at a single encounter is sufficiently large to change the perturbation in orbital phase by π before the next encounter. In such a case there is no correlation between the perturbations in longitudes at successive conjunctions. This implies that the system "forgets" the previous longitude and the motion is chaotic. In terms of the quantities used in Eqs. (9.95)–(9.97) this implies that chaos sets in when

$$\frac{4\pi \, \Delta\varepsilon_{n+1}}{3\varepsilon_1^2} \geq \pi \tag{9.148}$$

or

$$|\Delta\varepsilon| \geq \frac{3}{4}\varepsilon_1^2, \tag{9.149}$$

where we recall that ε is the relative difference in semi-major axis between the particle and the perturber. If we take the initial eccentricity to be zero and apply the condition after the first encounter, assuming that increasing $|\Delta\varepsilon|$ leads to increasing $|z| = e$, we have $|z_1| = 0$ and, from Eq. (9.96),

$$|z_2| = \frac{g}{\varepsilon_1^2}\frac{m_2}{m_1}. \tag{9.150}$$

Since

$$\varepsilon \, \Delta\varepsilon = \frac{2}{3}|z_2|^2 \tag{9.151}$$

the condition for chaos reduces to

$$\varepsilon_1 \leq \left(\frac{8}{9}g^2\right)^{1/7} \left(\frac{m_2}{m_1}\right)^{2/7}. \tag{9.152}$$

Putting in numerical values and making use of the fact that for the critical case $\varepsilon_1 = \Delta a_{\text{overlap}}/a'$, we obtain the result

$$\Delta a_{\text{overlap}} \approx 1.24 \mu_2^{2/7} a'. \tag{9.153}$$

This result is in good agreement with that originally derived by Wisdom (1980) using a different method.

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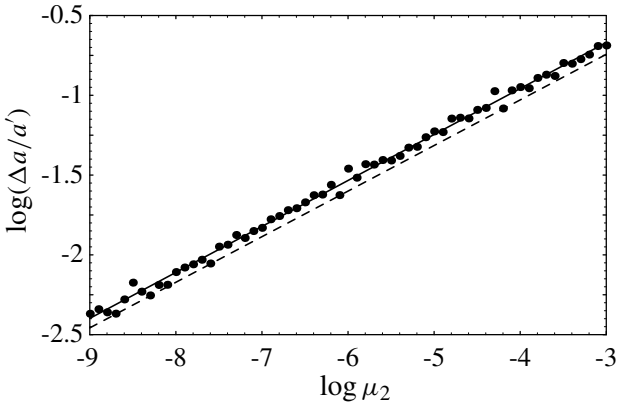


Fig. 9.23. The results of a series of numerical experiments to measure $\Delta a_{\text{overlap}}/a'$ using the encounter map devised by Duncan et al. (1989). The solid line is the best fit to the data. The dashed line denotes the theoretical result of Wisdom (1980).

We have checked the validity of this result by carrying out the same numerical experiment as performed by Duncan et al. (1989). This involved starting test particles on circular orbits at different relative separations $\Delta a/a'$ from the circular orbit of the perturber. The last value of $\Delta a/a'$ for which the orbit did not achieve an eccentricity $e > \Delta a/a'$ within 10^3 encounters was noted for a range of values of μ_2. The results are plotted in Fig. 9.23. The solid straight line corresponds to the best-fit solution of the form $\Delta a/a' = 1.57\mu_2^{0.286}$ and the dashed line denotes Wisdom's relation. Although there is good agreement between the slopes of each line, the values of $\Delta a/a'$ from the numerical experiment are slightly larger. In any case the Wisdom result seems to provide a lower bound on the numerical results.

9.7 The Rotation of Hyperion

We first introduced the concept of resonance in the context of spin–orbit coupling between a satellite and planet. In Sect. 5.4 we discussed the equation of motion governing the orientation of a satellite and showed how in certain circumstances we can obtain approximate analytical solutions to the motion. In Fig. 5.17b we plotted the values of $\dot{\theta}/n$ and θ (where θ is the orientation angle, $\dot{\theta}$ is its time derivative, and n is the mean motion) for a number of different starting conditions for a slightly nonspherical satellite moving in orbit with eccentricity $e = 0.1$. We noted the "fuzzy" appearance of some of the points near separatrices. These we can now explain as being due to a small chaotic zone. We were also able to derive approximate analytical solutions that mimicked the numerical results (see Figs. 5.17a), although not the chaos.

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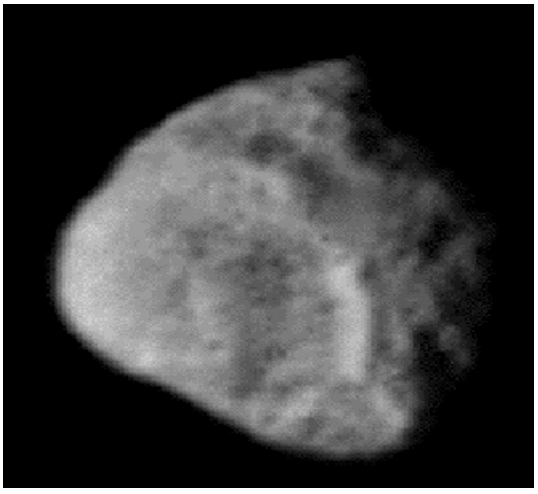


Fig. 9.24. A *Voyager 2* image of the saturnian satellite Hyperion showing its irregular shape. The longest dimension of the satellite is $\sim$ 350 km.

As pointed out in Chapter 5, most natural satellites in the solar system are in synchronous rotation, keeping one face pointed towards the planet. The timescale, T_{despin}, required for most satellites to settle into this configuration is considerably less than the age of the solar system (from Table 5.2 the value for the Moon is 3×10^7 y). However, it was shown in Sect. 5.5 that the value of T_{despin} is inversely proportional to the magnitude of the tidal torque, N_{s}, which is itself a strong function of the size of the satellite and inversely proportional to the sixth power of its distance from the planet (see Eq. (5.2)). Clearly, small satellites orbiting far from the parent planet may not have had time to evolve into a spin–orbit resonance. The saturnian satellite Hyperion has an unusual shape (see Fig. 9.24) with approximate radial dimensions of 175 km $\times$ 120 km $\times$ 100 km. Hyperion's orbital eccentricity is 0.1 and it has an orbital radius of 24.5 Saturn radii making it one of the most distant satellites of Saturn. Therefore we would expect such a small object at this distance to have a large value of T_{despin}.

Prior to the *Voyager* encounters with the saturnian system in 1980 and 1981, calculations of the tidal despinning timescales for the satellites suggested that Hyperion was a satellite that may not have had sufficient time to despin to a synchronous state. Initial observations of the light curve of Hyperion by Thomas et al. (1984) suggested that the *Voyager 1* and *2* data were best fitted by a spin period of 13 d. Since Hyperion's orbital period is 21.3 d, this implied that the rotation was nonsynchronous. The data also suggested that the spin axis was parallel to the orbital plane. To derive the spin period Thomas et al. (1984) had assumed a fixed rotation rate over a 61 d interval during the *Voyager 2*

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encounter. They also claimed that the derived period was consistent with the *Voyager 1* observations some 220 d earlier.

However, a paper by Wisdom, Peale & Mignard (1984) suggested a more unusual scenario. They carried out numerical and analytical studies of the solution to the differential equation describing the variation of θ given by Eq. (5.56). In the case of Hyperion the *Voyager* observations of its unusual shape suggested that a more appropriate value was $\omega_0 = 0.89 \pm 0.22$ (see Eq. (5.85)), assuming uniform density. This, combined with the large eccentricity of the orbit ($e = 0.1$), implied that not only was Hyperion not trapped in a spin–orbit resonance, but also that its rotation was *chaotic* and underwent essentially random but nevertheless deterministic changes. Since an averaging method is not appropriate for Hyperion, it is necessary to carry out a numerical integration. A surface of section for $e = 0.1$ and $\omega_0 = 0.89$ using values appropriate for Hyperion, is shown in Fig. 9.25. This shows the results of integrating a single trajectory for 20,000 orbital periods. Unless Hyperion happened to be trapped in one of the islands corresponding to resonant motion, the circumstantial evidence for a chaotic rotation is convincing. Note that by following the evolution of a single chaotic trajectory we can help to define the locations of the remaining regular regions; the same technique was employed to produce Fig. 9.15b for a completely different system. The result should be compared with Fig. 5.8 where seven different trajectories are shown and regular motion dominates.

The surface of section shown in Fig. 9.25 is similar to that found by Wisdom et al. and suggests that a large chaotic zone surrounds the $p = 1/2$ and $p = 2$

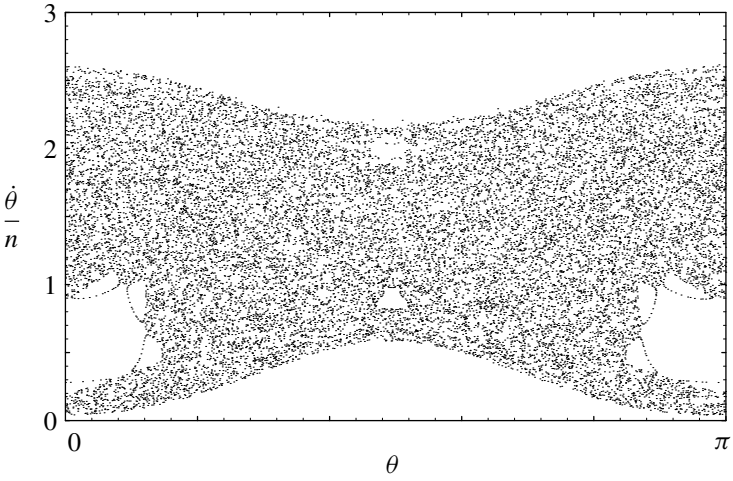


Fig. 9.25. A surface of section plot for a single trajectory in the $(\theta, \dot{\theta}/n)$ coordinate system using $\omega = 0.89$ and $e = 0.1$. The starting value is $\theta = 0.0$ and $\dot{\theta}/n = 0.28$ and the trajectory is followed for 20,000 orbital periods. (Adapted from Murray 1998.)

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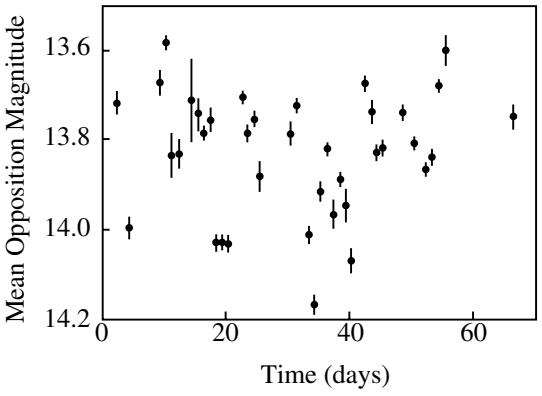


Fig. 9.26. Ground-based observations of Hyperion's light curve obtained by Klavetter (1989). Although the amplitude is large, there is no obvious periodicity and this is consistent with chaotic rotation.

spin states and also that libration in the $p = 3/2$ state is not possible. Wisdom et al. went on to show that the satellite was also likely to be *attitude unstable* such that it was tumbling in space. Ground-based observations by Klavetter (1989) appear to confirm these results (see Fig. 9.26).

It is interesting to note that the chaotic nature of Hyperion's rotation is a consequence both of its unusual shape and its large orbital eccentricity, although it is primarily due to the former. We have already seen that, left to itself, the tides raised on a satellite by a planet act to damp the eccentricity of the satellite's orbit. However, in the case of Hyperion the eccentricity cannot damp since it is a forced eccentricity due to its 4:3 orbit–orbit resonance with the nearby massive satellite Titan. Therefore, while the orbit of Hyperion appears to be stable because of the 4:3 resonance with Titan, its rotation is chaotic because of the competing effects of a variety of spin–orbit resonances.

However, the whole question of the rotation of Hyperion has still to be resolved. According to the work of Black et al. (1995), based on an improved shape model and rotation rate for Hyperion, although the spin state may be technically chaotic, the satellite can give the impression of regular rotation for long periods of time.

It is entirely likely that other irregularly shaped satellites have experienced chaotic rotation at some time in their history (Wisdom 1987a,b). In order for a satellite to get captured into spin–orbit resonance it has to cross chaotic separatrices and this is certain to have induced an episode of chaotic rotation. This could also have occurred if the satellite suffered a large impact that altered its rotational properties. Such episodes could also have produced significant internal heating and resurfacing events in some satellites. The Martian moon Phobos

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and the uranian moon Miranda have been mentioned as possible candidates for this process (see Wisdom 1987a,b and Dermott et al. 1988).

9.8 The Kirkwood Gaps

There are more than 10,000 asteroids with accurately determined orbits and the vast majority of these have semi-major axes between 2 and 4 AU, with a mean of $\langle a \rangle = 2.74 \pm 0.616$ AU. The total mass of the belt is $\sim 10^{-9}$ Earth masses, with $\sim 80\%$ of the mass contained in the three largest objects: Ceres, Pallas, and Vesta. The mean eccentricity is $\langle e \rangle = 0.148 \pm 0.086$ and the mean inclination is $\langle I \rangle = 8.58^\circ \pm 6.62^\circ$. A more useful quantity for comparison is the sine of the inclination; for the asteroid belt, $\langle \sin I \rangle = 0.148 \pm 0.111$. Therefore the mean and the standard deviation of the eccentricities and sine of the inclinations are comparable. Because the asteroids are small and numerous and undergo the significant gravitational effect of one major perturber (Jupiter) and several minor ones, they provide a natural laboratory for the study of dynamical processes in the solar system. We have already seen in Sect. 7.10 and Sect. 7.11 that there is good evidence for collisional evolution within the asteroid belt. This is not surprising since typical encounter velocities between asteroids are ~ 5 km s^{-1}, too large for accretion to occur.

The first asteroid was discovered by Guiseppe Piazzi on the first day of the nineteenth century. By the time Daniel Kirkwood carried out his study (Kirkwood 1867) of the distribution of the semi-major axes of asteroid orbits, he was able to make use of data on ninety-one objects. Kirkwood ordered the known asteroid orbits by semi-major axis and noted that the three widest separations occurred in the vicinity of the 9:4, 5:2, and 3:1 jovian resonances. He also calculated the location of the 7:2, 7:3, and 2:1 resonances but did not specifically associate them with gaps. However, Kirkwood was the first person to note that the distribution was nonrandom and was likely to have been influenced by jovian perturbations. He was of the opinion that *"As in the case of the perturbation of Saturn's ring by the interior satellites, the tendency of Jupiter's influence would be to form gaps or chasms in the primitive ring"* (Kirkwood 1867). It is clear that the inspiration for Kirkwood's study was the association of the Cassini division in Saturn's rings with the 2:1 resonance with Mimas. The mechanism that Kirkwood proposed involved the effect of repeated conjunctions. He believed that the likely buildup in the asteroid's eccentricity would lead to collision and coalescence with nearby objects causing depletion of material at the resonance.

As an historical footnote to the discovery of the gaps, we note that Kopal (1972) claimed that the existence of the gaps was first pointed out by a Prague astronomer, K. Hornstein, prior to the discovery by Kirkwood. We have found no documentary evidence to support Kopal's claim.

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9.8.1 Resonant Structure of the Asteroid Belt

Since Kirkwood's time the available data on the asteroid belt has grown by more than an order of magnitude. Now there are well-determined orbits for more than 10,000 asteroids. As well as orbital data there is a wealth on information available on asteroid colours, shapes, rotational properties, and, in the case of (243) Ida, satellites (see Fig. 2.6). All this information can be used to understand better the dynamical evolution of the asteroid belt.

Figure 9.27a shows a plot of the projected location of the asteroids in x–y space (the plane of the ecliptic) at 12h UTC on 18 December 1997. The limits on this plot exclude asteroids beyond $\sim$ 4 AU (for example, objects at the 4:3 and 1:1 jovian resonances) but there is still evidence of a well-defined ring of material. This can be highlighted by plotting a similar diagram with the semi-major axis and longitude of perihelion denoting the radial and angular positions respectively. Figure 9.27b shows clearly the resonant structure of the main belt, with a number of cleared "lanes" corresponding to the major Kirkwood gaps. The inner and outer edges of the belt are sharp and associated with the 4:1 and 2:1 resonances respectively (see Fig. 1.6), but the clustering at the 3:2 resonance (the Hilda group of asteroids) is clearly visible. The plot illustrates another feature of the asteroid belt – a clear bias in the angular distribution. Since the angular component of each point denotes the perihelion position of an asteroid, it is clear that there is a marked preference for longitudes of perihelion close to Jupiter's value (indicated by the arrow). This is simply a consequence of the fact that the precession rate of an asteroid dominated by jovian perturbations is at a

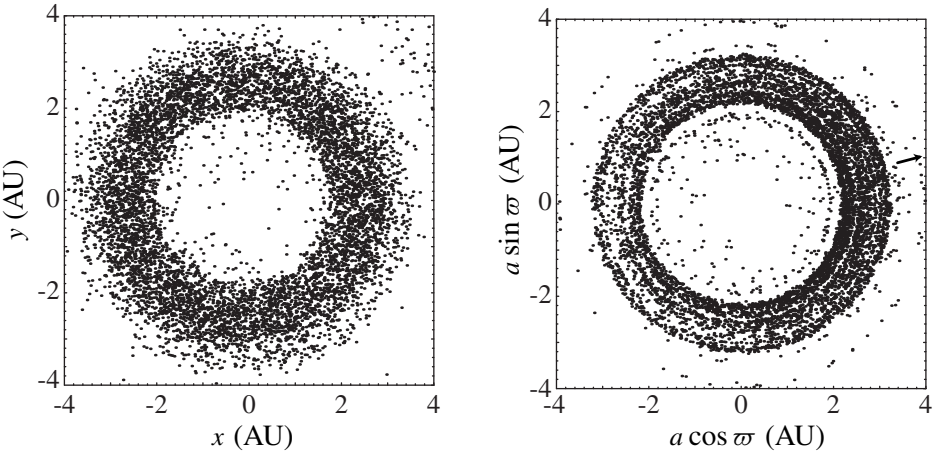


Fig. 9.27. (a) A plot of the x–y distribution of the known asteroids on 18 December 1997. (b) A plot of the $a\cos\varpi$–$a\sin\varpi$ distribution of the known asteroids using the same orbital data as in (a). Note the clear suggestion of a band of material corresponding to the main belt.

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minimum when the pericentres are aligned and a maximum when they differ by 180° (see Eq. (7.94)).

An important feature of the asteroid distribution is the decline in the number of asteroids beyond the 3:2 resonance (at 3.97 AU), apart from a small number of objects at the 4:3 resonance (at 4.29 AU). This absence of objects is also a resonant effect, caused by the overlapping of the sequence of first-order resonances as Jupiter is approached. Application of the resonance overlap criterion discussed in Sect. 9.6 (see Wisdom 1980) shows that in the Sun–Jupiter case $p_{\text{overlap}} \approx 4$, corresponding to a chaotic zone extending 0.9 AU inside Jupiter's orbit. Therefore we would expect a cleared zone in the asteroid belt beyond 4.3 AU; this is in good agreement with observations (see Fig. 9.28).

A number of hypotheses for the origin of the resonant gaps have emerged since Kirkwood's original discovery. It has to be pointed out that any viable theory must explain not only the gaps at certain resonances but also the groupings at others. Greenberg & Scholl (1979) classified the theories into four main groups:

- *The statistical hypothesis.* This is the claim that the gaps are simply an illusion. Since the dynamical equations governing the motion of objects in resonance are similar to those of a pendulum, if an asteroid is in resonance it will spend most of its time at the extremes of its motion. Consequently, at a given time, statistically the asteroid is more likely to be at the extremes of its librational motion (i.e., at the edges of the resonance rather than in the middle). With the

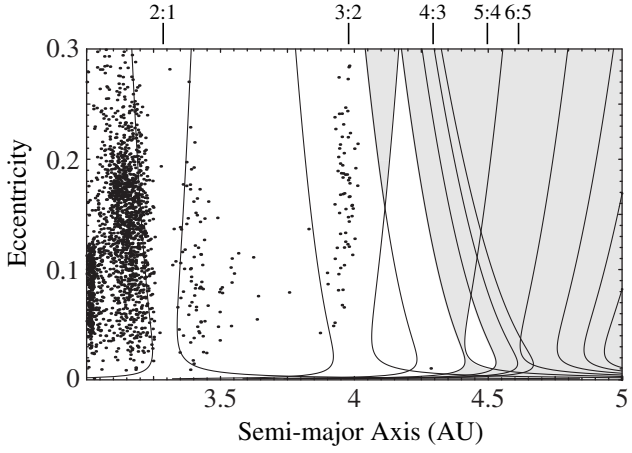


Fig. 9.28. The overlap of adjacent first-order resonances in the outer part of the asteroid belt. The curves denote the maximum libration widths in a–e space for all the first-order resonances between 3 and 5 AU. The points denote the a–e values of asteroids in this region. The nominal locations of the resonance centres are indicated for the 2:1, 3:2, 4:3, 5:4, and 6:5 resonances. The shaded areas denote regions of resonance overlap. Note the prominent gap at the 2:1 resonance and the grouping at the 3:2 resonance.

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advent of digital computers it became feasible to investigate this hypothesis by numerically integrating the equations of motion of known asteroids near the Kirkwood gaps. Apart from a few objects that were discovered to be librating, there was no indication that the gaps were illusory.

- *The collisional hypothesis.* This was originally proposed by Kirkwood. It suggests that the changes in orbital elements at a resonance cause the perturbed asteroids to collide with nearby objects. This could cause objects to accrete away from the gaps (unlikely since the the relative velocities would be too high) or a general breakdown of material due to collisional disruption. The latter can be checked by looking for evidence of a mass effect whereby the average mass of objects, as well as their number density, decreases as the resonance is approached.
- *The cosmogonic hypothesis.* Cosmogonic theories propose that the gaps are regions where asteroids failed to form or that they reflect processes that operated in the early stages of the formation of the solar system but have now ceased. For example, one idea is that *resonance sweeping* occurred when Jupiter's orbit started to expand, perhaps due to it orbiting a less massive solar nebula. Such a mechanism can be investigated using the resonance model discussed in Chapter 8. This may be able to explain the formation of some gaps, but not features such as the difference between the 2:1 and 3:2 resonances. Its effectiveness depends on unknown parameters such as the extent of orbital change in the early solar system.
- *The gravitational hypothesis.* According to this hypothesis the gaps can be understood in the context of the Sun–Jupiter–asteroid three-body problem. Research over the past two decades has shown that this is the most likely origin of the gaps and that chaotic motion is involved. However, the exact mechanism is not yet entirely understood.

Dermott & Murray (1983) showed that when the distribution of asteroid orbits is plotted in a–e space and the maximum libration widths of the strong resonances are superimposed, there is a good correspondence between the width of the gaps and the width of the libration zone (see Fig. 9.29). Note that the presence of an asteroid within the libration zone for a given resonance does not necessarily imply that it is in libration. This is because the data points for the asteroids are all plotted for the same epoch. A more correct comparison for an individual resonance would involve an integration of the orbit until the resonant angle for maximum libration had the appropriate value. Although Dermott & Murray provided no mechanism for the removal of asteroids, they concluded that the gravitational hypothesis was the most likely to be correct.

If the gaps can be understood in the context of the Sun–Jupiter–asteroid restricted problem then a simple, long-term integration of the equations should be sufficient to search for a removal mechanism. We have already carried out numerical investigations of this problem when we examined the surfaces of

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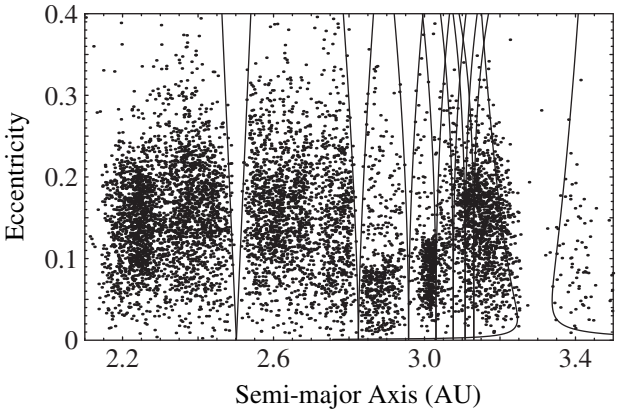


Fig. 9.29. The maximum libration widths in a–e space of the strong jovian resonances superimposed on the distribution of asteroids in the main belt. Note the correspondence between the widths of the gaps and the widths of the resonances.

section of the circular restricted problem for a mass ratio of 10^{-3} (see Sect. 9.4). In that case chaotic motion was detected. However, the central parts of each major resonance (i.e., each island in the surfaces of section) were regular in appearance. This leads one to believe that the Kirkwood gaps cannot be explained in the context of the planar, circular restricted problem alone.

Unfortunately long-term integrations of more complicated models (such as the elliptical and nonplanar problems) were almost impossible until the arrival of fast (and cheap) computers. The limited integrations that had been carried out previously did not reveal any dramatic behaviour in the motion of hypothetical asteroids at resonances. A major breakthrough occurred when Wisdom (1982) derived his algebraic mapping to study asteroid motion at the 3:1 resonance (see Sect. 9.5.2).

9.8.2 The 3:1 Resonance

In a series of papers Wisdom (1982, 1983, 1985a) provided convincing evidence of the role of chaos in the origin of the 3:1 Kirkwood gap. He used the map (Wisdom 1982) to show that for some starting conditions a test particle at the 3:1 resonance could give the appearance of regular behaviour for several thousand years and then undergo a large increase in its orbital eccentricity. At the location of the resonance ($a \approx 2.5$ AU) the asteroid would cross the orbit of Mars and eventually be removed by direct perturbations. Wisdom proposed that although the chaos in the motion is caused by Jupiter, it is Mars that actually removes the object. A sample plot of the chaotic evolution of a test particle at the 3:1 resonance is shown in Fig. 9.30. This was obtained using Wisdom's mapping technique.

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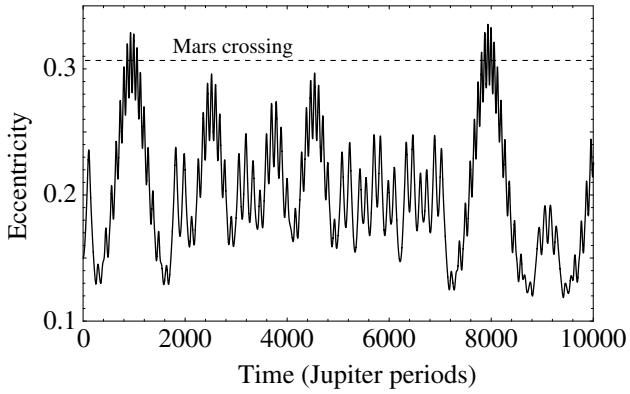


Fig. 9.30. The evolution of a test particle's eccentricity at the 3:1 resonance showing the chaotic nature of the orbit and how it can cross the orbit of Mars. The initial values were $a_0/a' = 0.481$, $e_0 = 0.15$, $\varpi_0 - \varpi' = 0$, and $3\lambda' - \lambda_0 = \pi$, corresponding to the chaotic, "representative" plane identified by Wisdom (1982). (Adapted from Murray 1998.)

In his initial paper Wisdom (1982) included the effect of secular perturbations on the orbit of Jupiter, as well as a full three-dimensional calculation. He went on to show (Wisdom 1983) that the observed chaos was not due to the secular evolution of Jupiter's orbit and that it occurred even in the planar case. However, Jupiter's eccentricity played a crucial role. Murray & Fox (1984) showed that the chaos was not an artifact of the map but was a genuine reflection of the type

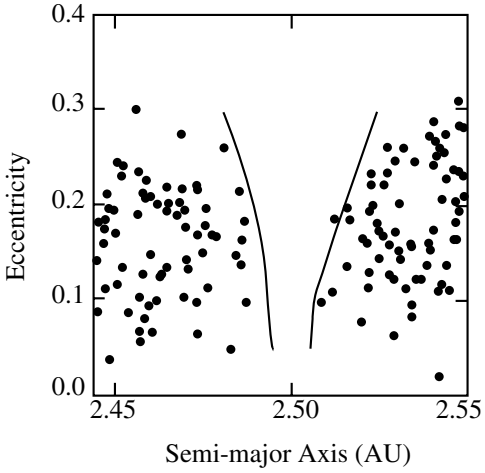


Fig. 9.31. The positions of asteroids at the 3:1 resonance compared with the extent of the chaotic region. (Adapted from Wisdom 1983.)

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of motion inherent in the averaged and full equations of motion. The excellent agreement between the width of the chaotic region and the width of the observed is illustrated in Fig. 9.31.

In further work Wisdom (1985a) provided a convincing analytical basis for the observed chaotic motion at the 3:1 resonance. He took the standard form of the Hamiltonian for the averaged, planar, restricted three-body problem at the 3:1 resonance:

$$\begin{aligned}\mathcal{H} = & -\frac{\mu_1^2}{2\Phi^2} - 3\Phi + \mu F(x^2+y^2) + e'\mu G x \\ & - \left[C(x^2-y^2) + e'Dx + e'^2 E\right]\cos\phi \\ & - \mu(C2xy + e'Dy)\sin\phi, \end{aligned} \tag{9.154}$$

where $\phi = \lambda - 3\lambda'$ and C, D, E, and F are constants with the other variables having their usual definitions. The Hamiltonian can also be written as

$$\begin{aligned}\mathcal{H} = & -\frac{\mu_1^2}{2\Phi^2} - 3\Phi + \mu F(x^2+y^2) + e'\mu G x \\ & - \mu\left[C(x^2-y^2) + e'Dx + e'^2 E\right]\cos\phi \\ & - \mu A(x,y)\cos[\phi - P(x,y)], \end{aligned} \tag{9.155}$$

where

$$A(x,y) = \sqrt{\left[C(x^2-y^2) + e'Dx + e'^2E\right]^2 + (2xyC + e'Dy)^2} \tag{9.156}$$

and

$$\tan P(x,y) = \frac{2xyC + e'Dy}{C(x^2-y^2) + e'Dx + e'^2E}. \tag{9.157}$$

Wisdom then assumes that x and y are "frozen" and can be treated as parameters in the equations of motion. The justification for this is that x and y depend on a, e, and ϖ, which are all slowly varying quantities that only change on resonant timescales. He defines a new λ by $\lambda = \phi - P(x,y) - \pi$ and its conjugate momentum $\Lambda = \Phi - \Phi_{\rm res}$, where $\Phi_{\rm res} = (\mu_1^2/3)^{1/3}$ is the nominal centre of the resonance defined by $\partial\mathcal{H}/\partial\Phi = 0$. Expanding $\mathcal{H}$ about $\Phi_{\rm res}$ and keeping only the quadratic terms gives a Hamiltonian $\mathcal{H}'$ governing the variations in λ and Λ:

$$\mathcal{H}' = \frac{1}{2}\alpha\Lambda^2 + \mu A\cos\phi, \tag{9.158}$$

where $\alpha = -3\mu_1^2/\Phi_{\rm res}^4 < 0$. This is another variation of the Hamiltonian of the simple pendulum (see Sect. 8.6). As before, the form of the solution depends on the value of $\mathcal{H}'$ and this, in turn, depends on the starting conditions. Three situations can arise: (i) If $\mathcal{H}' < -A$ the angle λ circulates, (ii) if $-A < \mathcal{H}' < A$ then λ librates about zero, and (iii) if $\mathcal{H}' = A$ then the motion is on the separatrix.

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In the case of libration the solution for λ is

$$\lambda = 4\sum_{n=1}^{\infty} \frac{\sin(2n-1)\omega t}{(2n-1)\cosh[(n-1/2)\pi K'/K]}, \tag{9.159}$$

where $K(k)$ and $K'(k)$ are the complete elliptical integrals of the first kind defined by

$$K(k) = \int_0^{\pi/2} (1 - k\sin^2\theta)^{-1/2} \mathrm{d}\theta \tag{9.160}$$

and

$$K'(k) = K(1-k) = \int_0^{\pi/2} \left[1 - (1-k)\sin^2\theta\right]^{-1/2} \mathrm{d}\theta \tag{9.161}$$

with modulus $k_\mathrm{L} = \sqrt{(\mu A - \mathcal{H}')/2\mu A}$ and $\omega = \pi\omega_0/2K$. The frequency ω_0 is the frequency of the small-amplitude oscillations given by $\omega_0 = \sqrt{-\alpha\mu A}$.

In the case of circulation the solution for λ is

$$\lambda = \omega t + 2\sum_{n=1}^{\infty} \frac{\sin n\omega t}{n\cosh(n\pi K'/K)}, \tag{9.162}$$

with $\omega = \pi\omega_\mathrm{C}/K$ and $\omega_\mathrm{C} = [-\alpha(\mu A - \mathcal{H}')/2]^{1/2}$. In this case the elliptic modulus is $k_\mathrm{C} = [2\mu A/(\mu A - \mathcal{H}')]^{1/2}$.

Using the Hamiltonian in Eq. (9.155) the equations of motion can be written

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -2\mu F y - \mu\frac{\partial A}{\partial y}\cos\lambda - \mu A\frac{\partial P}{\partial y}\sin\lambda, \tag{9.163}$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2\mu F x + e'\mu G + \mu\frac{\partial A}{\partial x}\cos\lambda + \mu A\frac{\partial P}{\partial x}\sin\lambda, \tag{9.164}$$

$$\frac{\mathrm{d}\lambda}{\mathrm{d}t} = \alpha\Lambda - \frac{\partial P}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\partial P}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}, \tag{9.165}$$

$$\frac{\mathrm{d}\Lambda}{\mathrm{d}t} = \mu A\sin\lambda. \tag{9.166}$$

From Eqs. (9.163) and (9.164) the timescale for significant evolution of x and y is proportional to $\mu^{-1/2}$. In a sense only the averaged effect of the rapidly oscillating terms containing $\cos\lambda$ and $\sin\lambda$ will contribute to the large variations of x and y. Wisdom suggested splitting x and y into their long- and short-period parts by writing $x = \bar{x} + \xi$ and $y = \bar{y} + \eta$ with the resulting equations of motion,

$$\frac{\mathrm{d}\bar{x}}{\mathrm{d}t} = -2\mu F\bar{y} - \mu\frac{\partial A(\bar{x},\bar{y})}{\partial\bar{y}}\langle\cos\lambda\rangle - \mu A(\bar{x},\bar{y})\frac{\partial P}{\partial\bar{y}}\langle\sin\lambda\rangle, \tag{9.167}$$

$$\frac{\mathrm{d}\bar{y}}{\mathrm{d}t} = 2\mu F\bar{x} + e'\mu G + \mu\frac{\partial A(\bar{x},\bar{y})}{\partial\bar{x}}\langle\cos\lambda\rangle + \mu A(\bar{x},\bar{y})\frac{\partial P}{\partial\bar{x}}\langle\sin\lambda\rangle, \tag{9.170}$$

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where the angled brackets indicate an average over one oscillation period of λ. These averages are

$$\langle \cos\lambda \rangle = \frac{1}{T}\int_0^T \cos\lambda \, \mathrm{d}t = \frac{2E(k_\mathrm{L})}{K(k_\mathrm{L})} - 1, \tag{9.171}$$

$$\langle \sin\lambda \rangle = \frac{1}{T}\int_0^T \sin\lambda \, \mathrm{d}t = 0 \tag{9.172}$$

for libration and

$$\langle \cos\lambda \rangle = \frac{1}{T}\int_0^T \cos\lambda \, \mathrm{d}t = \frac{2E(k_\mathrm{C})}{k_\mathrm{C}^2 K(k_\mathrm{C})} + 1 - \frac{2}{k_\mathrm{C}^2}, \tag{9.173}$$

$$\langle \sin\lambda \rangle = \frac{1}{T}\int_0^T \sin\lambda \, \mathrm{d}t = 0 \tag{9.174}$$

for circulation. In each case $E(k)$ is the complete elliptical integral of the second kind defined by

$$E(k) = \int_0^{\pi/2} (1 - k\sin^2\theta)^{1/2} \mathrm{d}\theta. \tag{9.175}$$

The equations of motion for x and y are then

$$\frac{\mathrm{d}\bar{x}}{\mathrm{d}t} = -2\mu F\bar{y} - \mu\frac{\partial A(\bar{x},\bar{y})}{\partial \bar{y}}\langle\cos\lambda\rangle, \tag{9.176}$$

$$\frac{\mathrm{d}\bar{y}}{\mathrm{d}t} = 2\mu F\bar{x} + e'\mu G + \mu\frac{\partial A(\bar{x},\bar{y})}{\partial \bar{x}}\langle\cos\lambda\rangle. \tag{9.177}$$

These equations effectively describe the motion of the guiding centre of trajectories at the 3:1 resonance. Wisdom used plots in $\bar{x}$–$\bar{y}$ space to illustrate the origin of the chaotic motion at the resonance. In particular he showed how certain starting conditions close to separatrices can give the impression of regular motion for long periods of time, before displaying sudden increases in eccentricity.

Meteorites are believed to be the fragments of material broken up by collisions within the asteroid belt, and there is certainly a good correlation between the reflectance properties of meteorite samples and asteroid spectra. The length of time taken for the meteorite to reach the Earth can be estimated from the *cosmic ray exposure age* of a sample. This is the length of time that the meteorite has been exposed to the general cosmic ray background of the solar system. Prior to the work of Wisdom (1985b) there had been a problem reconciling known exposure ages of $\sim 20 \times 10^6$ y of the ordinary chondrite class of meteorites with possible delivery mechanisms based on planetary perturbations. Wisdom (1985b) showed that the 3:1 resonance was the most likely source of meteorites. When the perturbations of the other planets are included, he showed that it is possible for objects at the 3:1 resonance to get eccentricities as large as 0.6, in which case they can cross the orbit of the Earth. At the same time Wetherill

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(1985) had shown that there had to be a source for the ordinary chondrites near 2.5 AU.

9.8.3 Other Resonances

The structure of the phase space at other jovian resonances has been investigated using maps, as well as numerical and analytical approaches. Murray (1986) derived a map for first-order resonances and used it to study the 2:1 and 3:2 jovian resonances. However, there is a severe problem with the convergence of the disturbing function at these resonances for any but the lowest values of e. The results of Murray's map were only valid for low e at the 2:1 resonance and probably invalid at the 3:2 resonance. In any case Murray failed to demonstrate any fundamental difference between the two resonances that could account for a concentration of objects at the 3:2 and a lack of them at the 2:1. However, making use of a computer called the Digital Orrery, which was built specifically for this purpose, Wisdom (1987a) showed that chaos is again responsible. From the results of numerical integrations he deduced the presence of a large chaotic zone at the centre of the 2:1 resonance and the absence of such a zone at the 3:2 resonance.

At this stage it is important to point out that not all chaotic orbits result in gross instabilities that lead to planet-crossing trajectories. The study by Milani & Nobili (1992) of the motion of asteroid (522) Helga showed that it has a measurable Lyapounov exponent of $1/6600\ \mathrm{y}^{-1}$; yet numerical integrations showed no sign of it having a close approach to Jupiter or any of the other planets. This is another example of the phenomenon of bounded chaos mentioned in Sect. 9.3.3. In the case of (522) Helga the chaos may be connected with its proximity to the 12:7 jovian resonance. Murray & Holman (1997) showed that Helga's evolution is governed by a random walk.

It seems likely that the dynamical structure of the asteroid belt has been shaped by planetary perturbations, most notably those from Jupiter. Therefore the current, evolved population of asteroids must represent those orbits from the original population that have managed to survive the cumulative effects of chaos and collisions over the age of the solar system.

A general study of the dynamical evolution of planet-crossing asteroids was undertaken by Milani et al. (1989). They carried out numerical integrations of 410 asteroids for 2×10^5 y, including the gravitational perturbations from all the planets except Mercury and Pluto. Their results showed that the long-term motion of some of these asteroids is remarkably complex. For example, a plot of the semi-major axis of the Earth-crossing asteroid (1620) Geographos shows that it undergoes temporary trapping in a number of high-order resonances with the Earth in the course of its chaotic evolution (see Fig. 9.32). A study of high-order, jovian mean motion resonances in the outer asteroid belt (Holman & Murray 1996) has shown that even weak resonances can give rise to significant chaotic zones.

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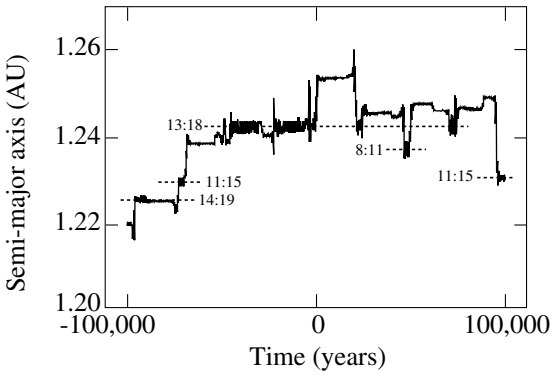


Fig. 9.32. The orbital evolution of the semi-major axis of (1620) Geographos. (Adapted from Milani et al. 1989.)

The advances in high-speed, cheap digital computers and the use of a variety of mapping techniques have led to many developments in the study of the long-term motion of asteroids and comets under planetary perturbations. The view that the evolution of such objects is dominated by a single planet (Jupiter in the case of the asteroids) or even a single resonance has now been found to be too narrow. Wisdom (1985b) had already shown that once perturbations from the other planets are included objects at the 3:1 jovian resonance could achieve eccentricities > 0.6 and so could become Earth-crossing. This strongly suggested that secular interactions between the planets also played a role in determining long-term stability. The results of many recent integrations have shown that secular resonances (see Sect. 7.11 and Fig. 7.19) rather than mean motion resonances can be the major perturbing influences (see, for example, Moons & Morbidelli 1995, Froeschlé et al. 1995). In fact, as shown by Gladman et al. (1997), the dynamical evolution of objects injected into mean motion resonances can be extremely complicated.

9.9 The Neptune–Pluto System

Pluto has the largest eccentricity ($e = 0.25$) of any planet and a plot of its orbital path appears to show an intersection with the orbit of Neptune (see Fig. 2.15b). However, plots that show the projection of Pluto's orbit in the plane of the ecliptic fail to bring out the effect of the inclination. Figure 9.33 shows the orbits and positions of Neptune and Pluto in 1989 when Pluto was at perihelion. The dashed curves in the plot indicate those portions of the orbit which lie below the plane of the ecliptic and the straight lines are the lines of nodes for each orbit. Neptune moves on a near-circular orbit with an inclination of $< 2^\circ$ (see Table

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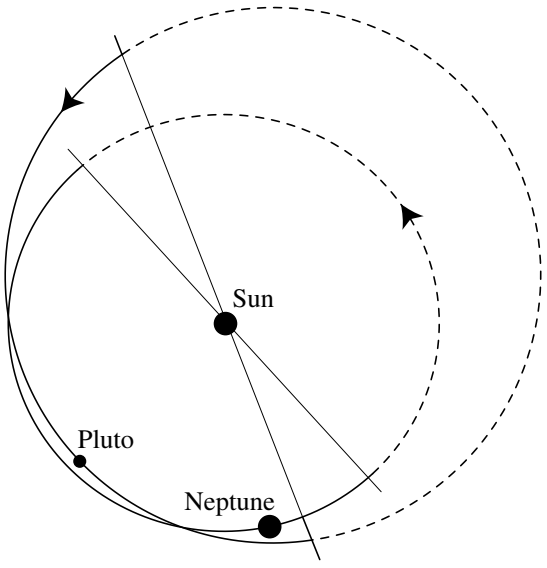


Fig. 9.33. The orbital configurations of Neptune and Pluto when Pluto was at perihelion in 1989. The straight lines denote the lines of nodes of each orbit. The dashed part of each orbit denotes the parts of the orbit below the plane of the ecliptic.

A1.2) and so it is always within 1 AU of the ecliptic. In contrast, Pluto has an eccentric orbit with a large inclination (17°), which implies significant motion out of the ecliptic. Furthermore, Neptune and Pluto are in a 3:2 resonance and the combination of effects means that when Neptune and Pluto are at conjunction, a close approach is avoided.

In Fig. 9.34 we plot the separation in AU between Neptune and Pluto over a timescale of some 1,000 years, centred on 1989. This shows that even though the orbits appear to intersect, the two planets never approach closer than ~ 20 AU, significantly larger than that suggested by Fig. 9.33. Note that this plot is based on two separate two-body problems and no mutual perturbations are included. However, it does illustrate that the geometry of the orbits is important in determining the closest approach distance.

Pluto was discovered by Clyde Tombaugh in 1930 and it has an orbital period of 248 y. The discovery of the Edgeworth–Kuiper belt (see, for example, Luu & Jewitt 1996) has shown that Pluto is just one member of a whole class of objects orbiting in this region of the solar system, some of which (the *Plutinos*; see Jewitt & Luu 1996) are also known to be in a 3:2 resonance with Neptune. However, none of the objects has yet to be observed over a complete orbital period. This implies that dynamical studies of possible resonances have to be accomplished by

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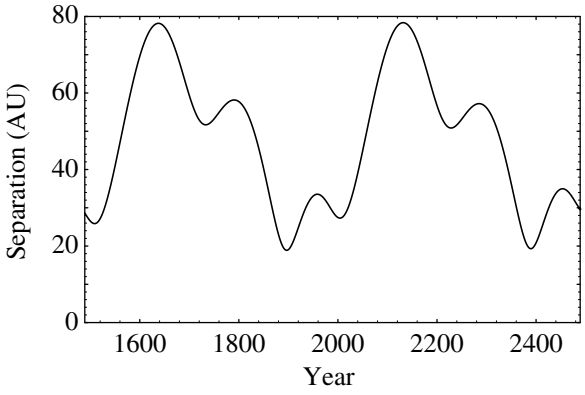


Fig. 9.34. The distance between Neptune and Pluto as a function of time based on individual two-body motions of the Sun–Neptune and Sun–Pluto systems.

numerical integration. The first such integration of Pluto's orbit was undertaken by Cohen & Hubbard (1965). On the basis of a backward integration covering 120,000 y they deduced that Pluto and Neptune are involved in a resonance with argument $\varphi = 3\lambda' - 2\lambda - \varpi'$, where λ and λ' denote the mean longitudes of Neptune and Pluto respectively and ϖ' is the longitude of perihelion of Pluto. Cohen & Hubbard showed that in the Neptune–Pluto system φ librates about 180° with an amplitude of $\sim 76^\circ$ and a period of 19,670 y. Figure 9.35 shows the approximate path of Pluto in a frame rotating with the radius vector from the Sun to Neptune.

Sussman & Wisdom (1988) carried out a 845 million year integration of the outer planets using the Digital Orrery and showed that the motion of Pluto is chaotic. In a previous integration covering 200 million years, Applegate et al.

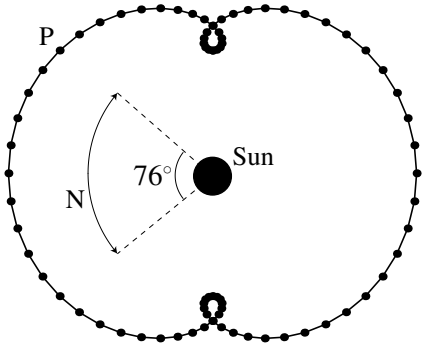


Fig. 9.35. The path of Pluto (P) in a frame rotating with the radius vector from the Sun to Neptune (N). The points along the path are placed at equal time intervals. The arc denotes the 76° amplitude of φ about 180°.

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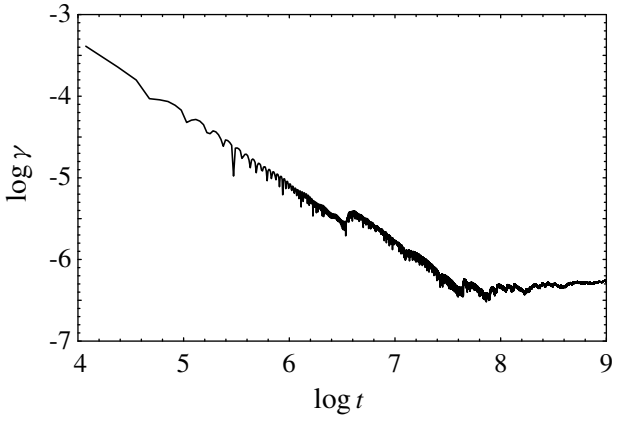


Fig. 9.36. Calculation of $\log \gamma$ as a function of $\log t$ for Pluto, where t is the time in years and γ is an estimate of Pluto's maximum Lyapounov exponent.

(1986) showed that Pluto's argument of perihelion is modulated with a period of 34 million years and $h' = e' \sin \varpi'$ has significant long-period variations with a period of 137 million years. The variation in h' is due to a near-commensurability between the frequency of circulation of Pluto's longitude of ascending node and one of the eigenfrequencies of the system. The longer integration by Sussman & Wisdom (1988) confirmed these periods and also showed evidence for a component with a period of $\sim$ 600 million years. Sussman & Wisdom also calculated a maximum Lyapounov exponent, γ, for the system and found that $\gamma \approx 10^{-7.3}$ y^{-1}, implying an e-folding timescale for exponential divergence of $\sim$ 20 million years. Figure 9.36 (cf. Fig. 9.9) shows the results of a similar calculation based on the output of a numerical integration over 10^9 y. Notice that the slope of the plot is approximately -1 for $\sim 10^{7.5}$ y before γ tends to a fixed value (in this case a value larger than that found by Sussman & Wisdom). This demonstrates how a chaotic orbit can give the appearance of being regular for long time intervals before its chaotic nature becomes apparent. Despite the existence of chaos in Pluto's orbit, there is no evidence that it will ever leave the resonance with Neptune, even in integrations over timescales as long as the age of the solar system (Kinoshita & Nakai 1996). An understanding of the origin of this apparently stable, yet chaotic resonance may involve the orbital migration of Neptune in the early history of the solar system (Malhotra 1993).

9.10 The Stability of the Solar System

In our discussion of secular perturbation theory, we showed that the solar system is stable according to the Laplace–Lagrange theory. This is because our simplifying assumptions enabled us to derive an integrable solution to the problem

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of the secular changes in the orbits of N planets. However, we also pointed out that these assumptions (low e and I, no mean motion resonances, no effect of short-period terms) are not strictly valid for the planets. For example, we know that Jupiter and Saturn are in a 5:2 near-resonance. In some circumstances it is possible to modify the secular theory to take account of such effects, but it is important to realise that in such analytical approaches it is possible to neglect some potentially important interactions between the planets. If we go on to consider the possibility of near-resonances between the secular frequencies of the system the problem becomes even more difficult.

In recent years there have been two separate approaches to the problem, both of them numerical. The advent of fast, cheap computers has meant that it is now possible to carry out numerical integrations of the full equations of motion of the planetary orbits for times approaching the age of the solar system. A number of such projects were carried out in the 1980s, most involving integrations of the outer planets only (Jupiter, Saturn, Uranus, Neptune, and Pluto). These include the work by Kinoshita & Nakai (1984), Applegate et al. (1986), and Roy et al. (1988). The reason for the early focus on the orbits of the outer planets was the problem with timescales. Any numerical integration of the entire solar system would have to use a time step that was suitable for following the orbits of both Mercury and Pluto simultaneously. These time steps are determined by the orbital period of the innermost (i.e., fastest-moving) object. There are also significant problems with truncation error and the buildup of round-off error, although some of these can be alleviated with a suitable choice of time step (see Sussman & Wisdom 1988). As a test of their symplectic mapping, Wisdom & Holman (1991) re-did the calculation of Sussman & Wisdom (1988) and extended the integration of the outer solar system to one billion years.

An alternative approach to full integration (with or without a mapping technique) is to generate a system of averaged equations using appropriate terms from the disturbing function. Such a system of equations is an approximation to the real system and requires significant algebraic calculations (usually performed by computer) before the numerical integration can begin. However, the equations can be integrated using a much larger time step because the short-period effects have been averaged out. For example, 40 d is a typical time step in a full integration of the outer planets whereas a typical value for an averaged system is 500 y. In his study of the evolution of all the planets over 10 million years, Laskar (1988) integrated the averaged equations. He showed that the inner planets are also chaotic and have maximum Lyapounov exponents of $\sim 10^{-6.7}$ y^{-1}, giving e-folding timescales of 5 million years.

All these numerical integrations show a consistent picture of planets moving on chaotic orbits. However, none show any sign of gross instabilities in the system for integrations over timescales comparable to the age of the solar system. Therefore, although the planets move on chaotic orbits the chaos is subtle. It implies that there are fundamental limits to our ability to predict the positions

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of the planets for long time intervals. If we had the starting positions and velocities of the planets with infinite precision and then had a perfect integrator, we would be able to calculate the future or past orbits. However, since any physical measurement has finite precision, there will always be an in-built error in the calculations. In a chaotic system this error will propagate exponentially. In the case of planets such as the Earth this means that we cannot predict the future position of the Earth in its orbit. For example, a maximum Lyapounov exponent of $10^{-6.7}$ means that an error of 1 cm in the position of the Earth today propagates such that we cannot predict its location 200 million years in the future.

Laskar (1994) extended his integrations by investigating the planetary orbits for 10 billion years into the past and 15 billion years into the future. These showed a close coupling of the Earth and Venus in terms of their variations of eccentricity and inclination. The results also showed dramatic, chaotic variations in the orbits of Mercury ($0.1 < e < 0.5$, $8^\circ < I < 21^\circ$) and Mars ($0.1 < e < 0.2$, $4^\circ < I < 10^\circ$). Although an eccentricity of 0.5 is not sufficient for Mercury to cross the orbit of Venus, it prompted Laskar (1994) to carry out further integrations, shifting the initial position of the Earth by 150 m. Some of the integrations showed that Mercury's eccentricity could reach a value $e \approx 1$ (i.e., near-hyperbolic) at epochs 3.5 billion years in the future or 6.6 billion years in the past. However, when assessing the significance of these results it is important to remember that the physical model of point (constant) mass objects orbiting in a conservative system may no longer be applicable when such timescales are considered.

With the possible exception of Mercury (see above), all integrations show that the planets remain close to their current orbits for timescales approaching a billion years or more, even though the orbits are technically chaotic. In this sense the solar system is stable and it is likely that motions of the planets represent another case of bounded chaos. As yet there is no analytical proof of the solar system's stability or the origin of the chaos. Analyses of the stability properties of other planetary systems are just beginning (see, for example, Holman et al. 1997).

Exercise Questions

9.1 Write a computer program to solve numerically the Cartesian equations of motion of the planar, circular, restricted three-body problem in the rotating reference frame (Eqs. (3.16) and (3.17)). Your program should take initial values of the orbital elements, convert them to Cartesian position and velocity in the rotating frame (choosing units such that the mean motion of the two masses is unity), solve the equations to find the new position and velocity at fixed time intervals, and convert them to orbital elements for output. Consider the motion of a test particle in the Sun–Jupiter system with $\mu_2 = 10^{-3}$, where Jupiter is always

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started at zero initial longitude and the unit of distance is the radius of Jupiter's orbit. The particle has initial semi-major axis $a_0 = 0.8$, eccentricity $e_0 = 0.4$, and longitude of perihelion $\varpi_0 = 295^\circ$. Numerically integrate the equations of motion for one Jupiter period using initial mean longitude $\lambda_0 = 293^\circ$ and $\lambda_0 = 293.3^\circ$ (see Fig. 9.1), giving the differences between the initial and final values of a, e, and ϖ in each case. By varying the value of λ_0 in steps of 0.1°, find the value of λ_0 that gives (i) the largest increase and (ii) the largest decrease in semi-major axis. Explain your results in terms of the geometry of the Jupiter encounter.

9.2 Write a computer program to solve numerically the Cartesian equations of motion of the planar, circular, restricted three-body problem in the rotating reference frame (Eqs. (3.16) and (3.17)) to produce a sequence of pairs of values of x and $\dot{x}$ whenever $y = 0$ with $\dot{y} > 0$. Use your program with $\mu_2 = 10^{-3}$ and initial values $\dot{x} = 0$ with $0.2 \leq x \leq 0.8$ (at intervals of 0.01) to produce a Poincaré surface of section for prograde, interior orbits with a Jacobi constant $C_J = 3.03$. Each integration should be followed until 500 points have been produced on the surface of section *or* the trajectory gets within a distance $\alpha = (\mu_2/3)^{1/3} \approx 0.07$ of the secondary mass *or* the radial distance exceeds 3. Identify any resonant "islands" that appear in the resulting surface of section.

9.3 Use the planar form of Wisdom's resonance map given in Sect. 9.5.2 to write a computer program to study the motion of asteroidal test particles at the 3:1 resonance with Jupiter. The program should take initial values of the semi-major axis (a), eccentricity (e), longitude of perihelion (ϖ), and mean longitude (λ) of the asteroid and the equivalent values a', e', ϖ', and λ' for Jupiter; calculate the initial values of the quantities x, y, Φ, and φ defined in Sect. 9.5.2; calculate the final values of these quantities after the seven steps of the map; and return the new elements for the asteroid one Jupiter period later, taking Jupiter's orbit to be fixed. Use the program with initial values $a/a' = 0.481$, $e = 0.15$, $e' = 0.048$, $\varpi = \varpi' = 0$, and $3\lambda' - \lambda = 180^\circ$ to plot the evolution of the asteroid's eccentricity over 10^4 Jupiter periods (cf. Fig. 9.30). Repeat the run with the same initial values of a, a', e', ϖ, ϖ', and $3\lambda' - \lambda = 180^\circ$ but using $e = 0.1500001$ to produce a similar plot of e as a function of time and comment on the difference between the two plots. Plot the difference in eccentricity between the two runs as a function of time and use this to estimate the maximum Lyapounov characteristic exponent.

9.4 The computer program written for Question 9.3 can be used to carry out a study of asteroid motion at the 3:1 jovian resonance. Take an array of starting conditions in the "representative" plane using $0.47 \leq a/a' \leq 0.49$ (at intervals of 0.001), $0 \leq e \leq 0.3$ (at intervals of 0.01), with $\varpi = \varpi' = 0$, $3\lambda' - \lambda = 180^\circ$, and $e' = 0.048$, where e' and ϖ' are taken to be constant. Follow each trajectory

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and plot those starting values of a and e that give rise to orbits with $e > 0.3$ (i.e., approximate Mars-crossing orbits) within 10^4 Jupiter periods.

9.5 Use the encounter map of Duncan et al. given in Eqs. (9.92)–(9.94) to write a computer program to study the orbital evolution of test particles due to successive conjunctions with Jupiter, where the orbit of Jupiter (mass ratio $m_2/m_1 = 9.545 \times 10^{-4}$ and semi-major axis $a' = 1$) and the initial orbit of the test particle are taken to be circular, and where the objects are initially at conjunction. For each initial value of the particle's semi-major axis a, starting at $a = 0.7$ with increments of 0.001, have the program follow the particle for 10^4 encounters. What is the last initial value of $\Delta a = a' - a$ for which the given starting condition did not achieve $e > \Delta a/a'$ within 10^4 encounters and how does this compare with the value predicted on the basis of the resonance overlap criterion?

9.6 Write a computer program to implement the Wisdom & Holman N-body map discussed in Sect. 9.5.4. Use the program to integrate the orbits of the planets for 50 years starting at the epoch of J2000 (JD 2451545.0) using the data given in Table A.2 and the masses given in Table A.4. Use Eq. (A.14) and Table A.3 to calculate the mean longitude of each planet at 100 day intervals for 50 years using a mapping interval $\Delta t = 5$ days. Compare these values with those obtained using the map with a mapping interval of (a) 1 day and (b) 10 days, listing the maximum difference over the 50 year interval in each case.

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